



Large-Scale Prediction of Curve Patch Replacement Using High-Frequency Data and Condition Matching

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Founder, VisioStack

DALLAS, TX



**Located in
Greenville, South
Carolina**

Founded in 2011

**Software Developers,
Data Scientists,
Mathematicians,
Railway Industry
Experts**



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Curve Patch is currently done on a year-by-year basis, using data from this year to guide capital planning for next year

Inaccurate, infrequent, and incomplete data sets hamper the ability to accurately predict when rail replacement is needed

Our goal is to fundamentally change how capital expenditure is calculated by extending the horizon of visibility to 3 to 5 years



North American customers by the numbers

More than 100,000 miles of track assets in the platform

Over **100,000** curves spanning **30,000** miles

Nearly **15 million miles** of data in the platform with over **3 million** of that being rail wear data

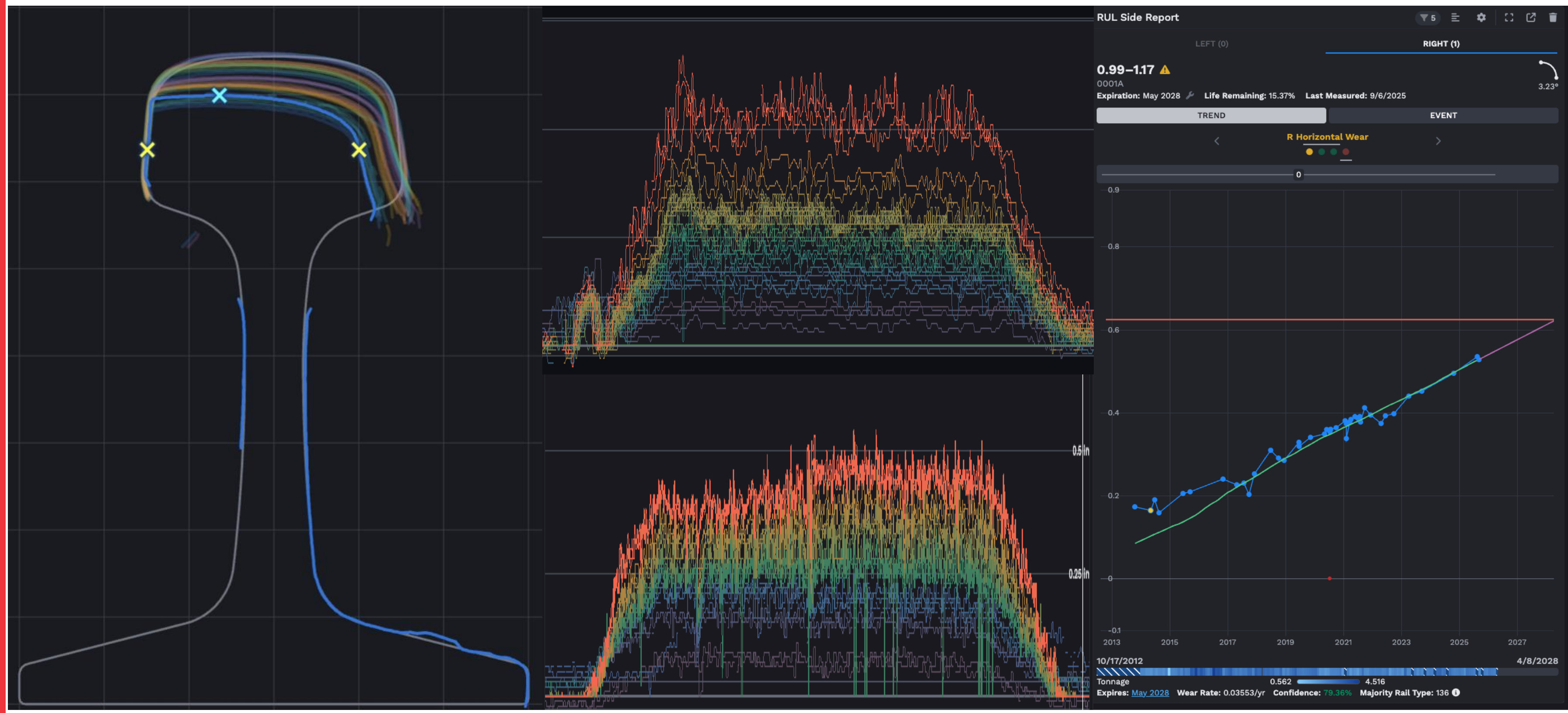
In 2026 our platform is being used to guide decisions covering over **1400 track miles** of rail replacement ¹

This represents over **1 billion dollars** in capital investment decisions

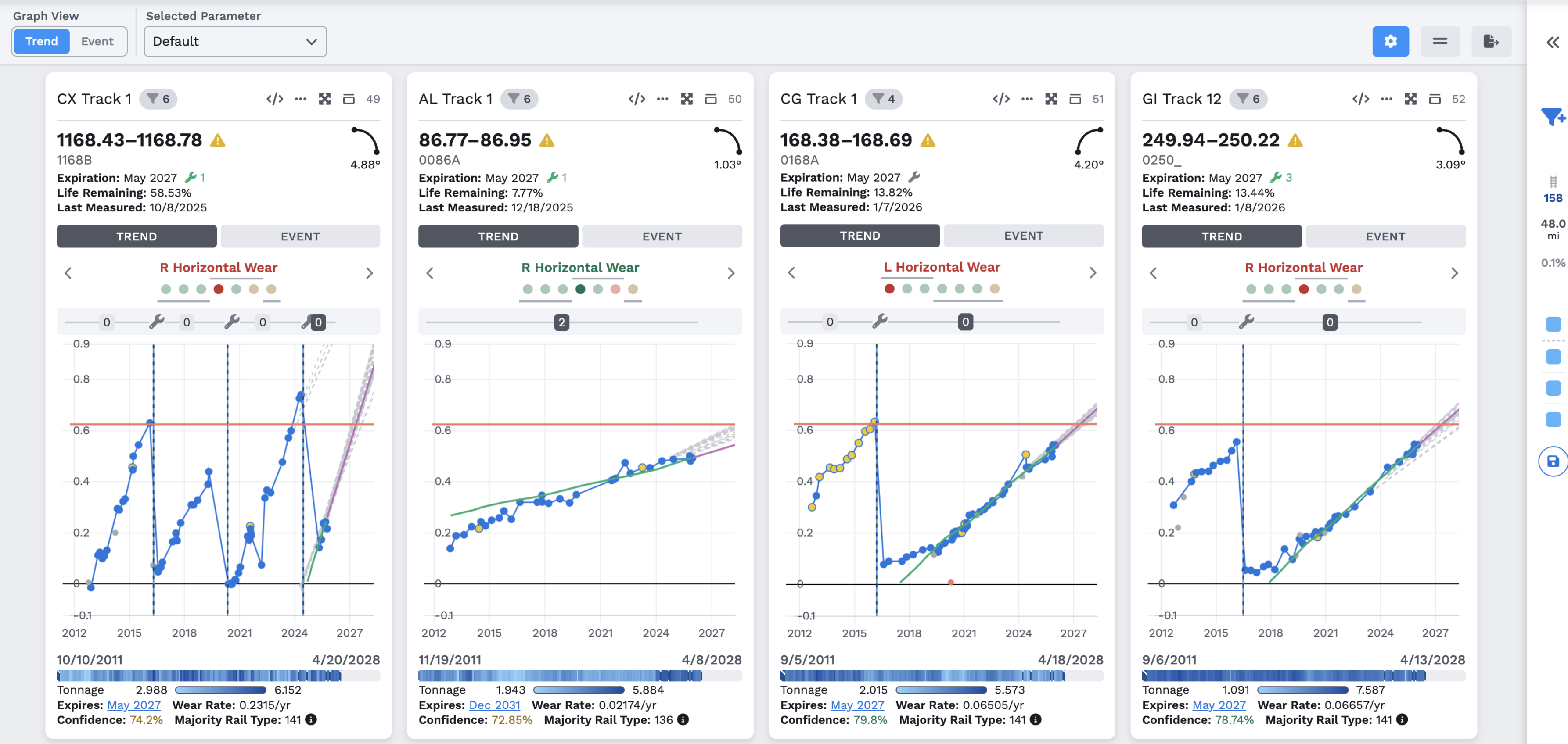
¹ Progressive Railroading 2026 MOW Spending report



Basic components of curve patch prediction



All curves predicted for a given year



AL Track 1

6

50

86.77–86.95

0086A

Expiration: May 2027

Life Remaining: 7.77%

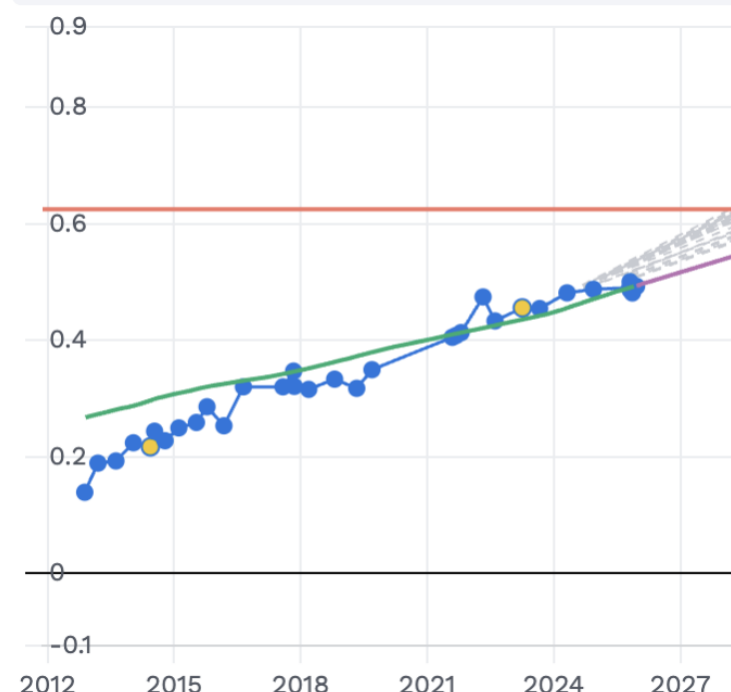
Last Measured: 12/18/2025

TREND

EVENT

R Horizontal Wear

2



11/19/2011

4/8/2028

Tonnage

1.943

5.884

Expires: Dec 2031

Wear Rate: 0.02174/yr

Confidence: 72.85%

Majority Rail Type: 136

CG Track 1

4

51

168.38–168.69

0168A

Expiration: May 2027

Life Remaining: 13.82%

Last Measured: 1/7/2026

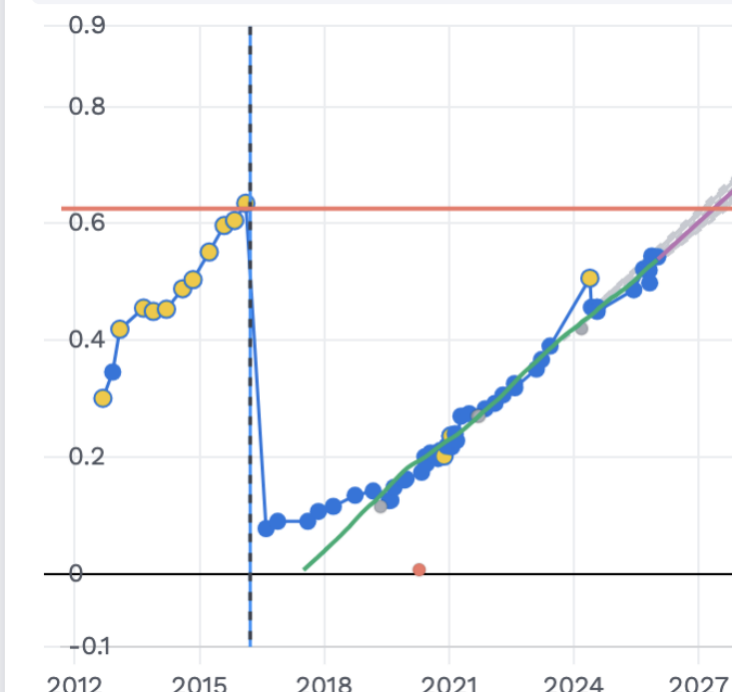
TREND

EVENT

L Horizontal Wear

0

0



9/5/2011

4/18/2028

Tonnage

2.015

5.573

Expires: May 2027

Wear Rate: 0.06505/yr

Confidence: 79.8%

Majority Rail Type: 141

GI Track 12

6

52

249.94–250.22

0250_

Expiration: May 2027

Life Remaining: 13.44%

Last Measured: 1/8/2026

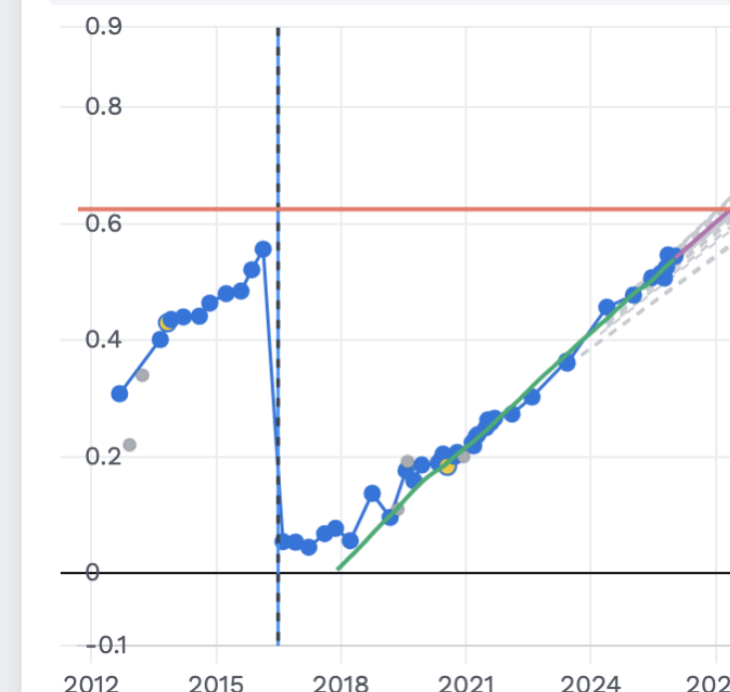
TREND

EVENT

R Horizontal Wear

0

0



9/6/2011

4/13/2028

Tonnage

1.091

7.587

Expires: May 2027

Wear Rate: 0.06657/yr

Confidence: 78.74%

Majority Rail Type: 141

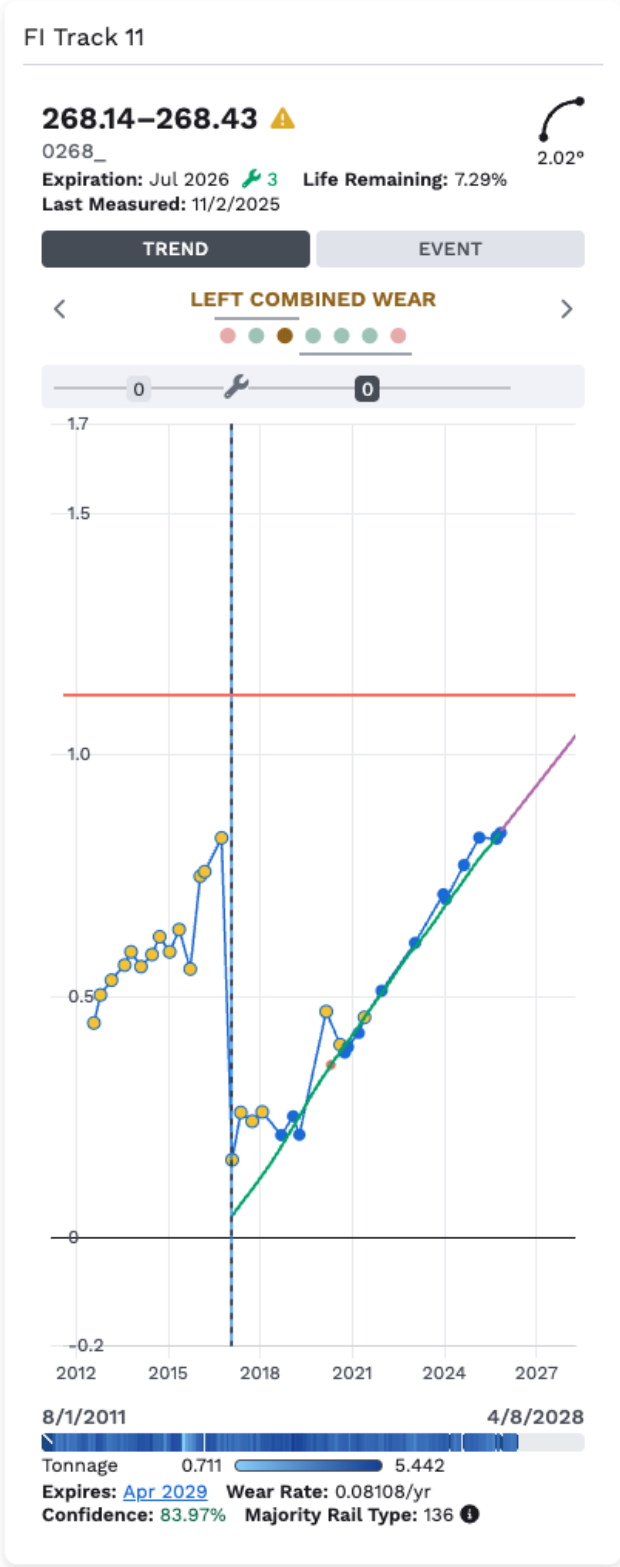
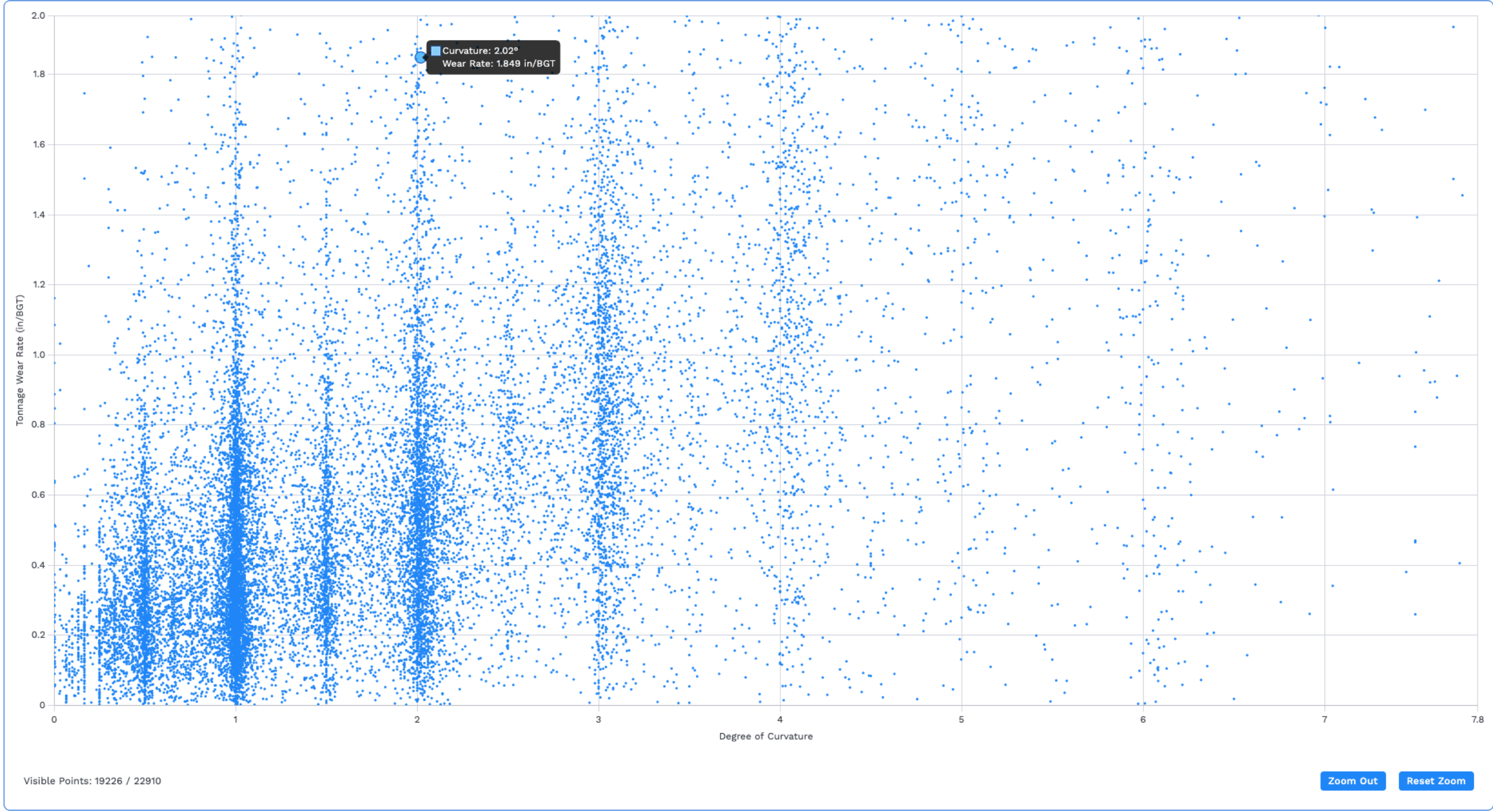
158

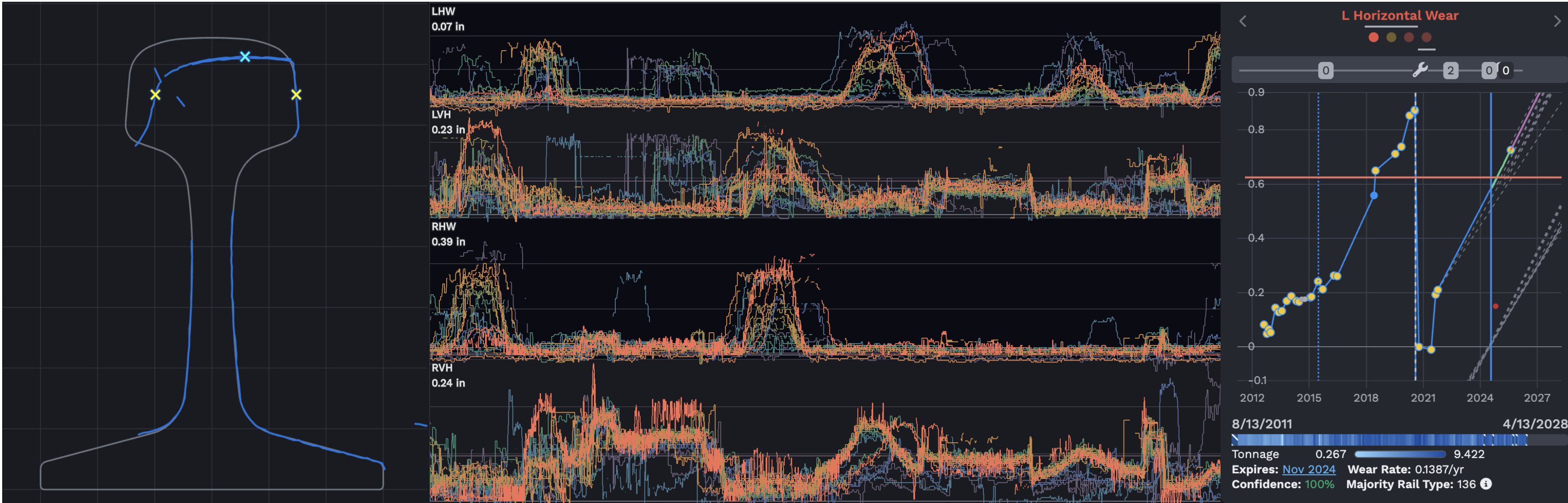
48.0 mi

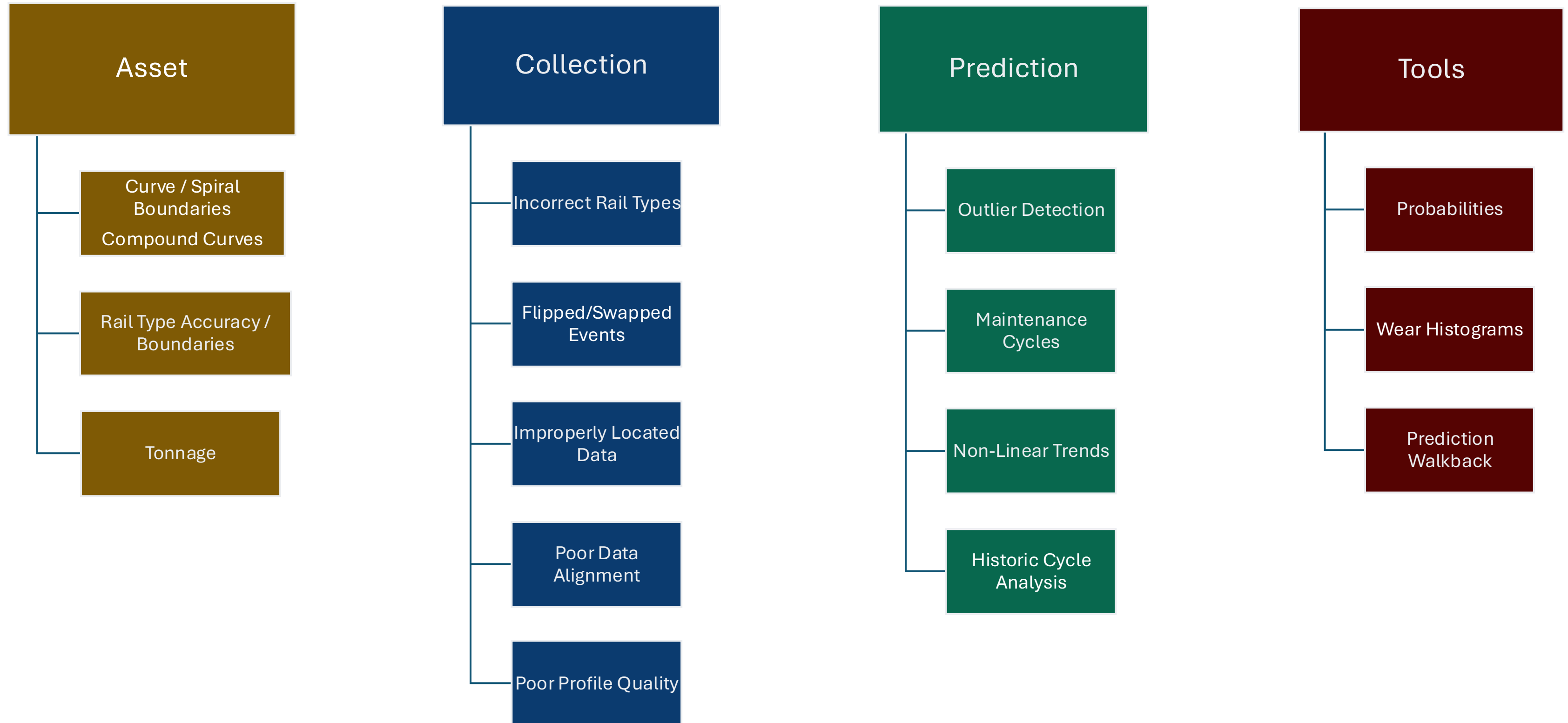
0.1%



Wear rates across an entire network

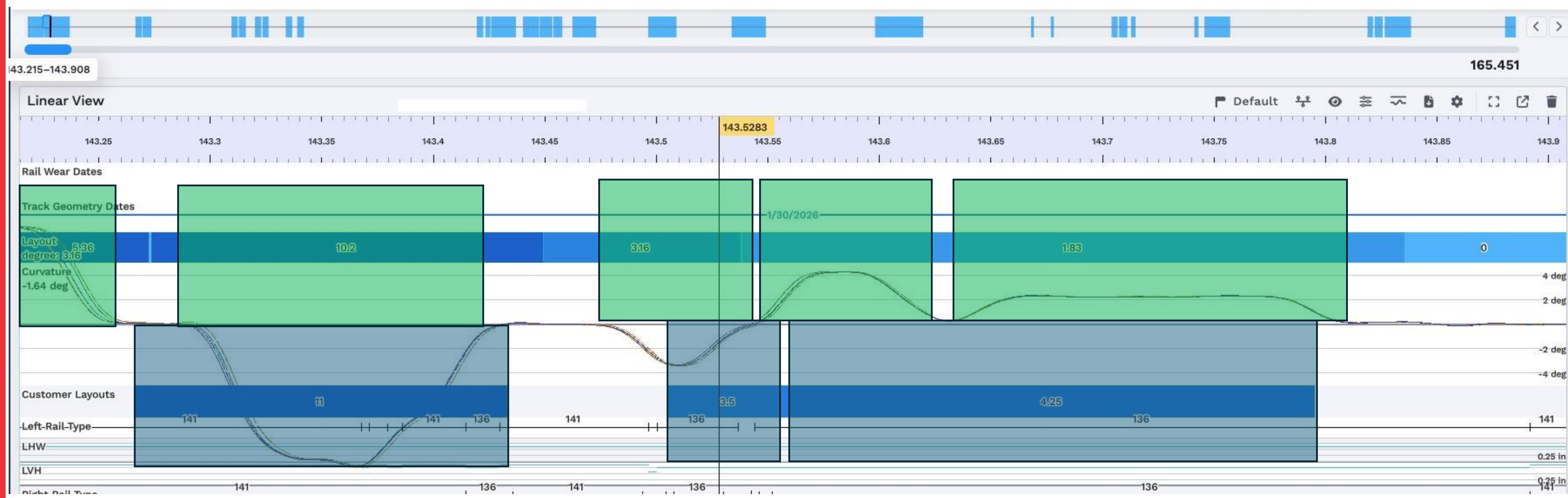






Having a well-defined asset is the foundation of a good prediction.

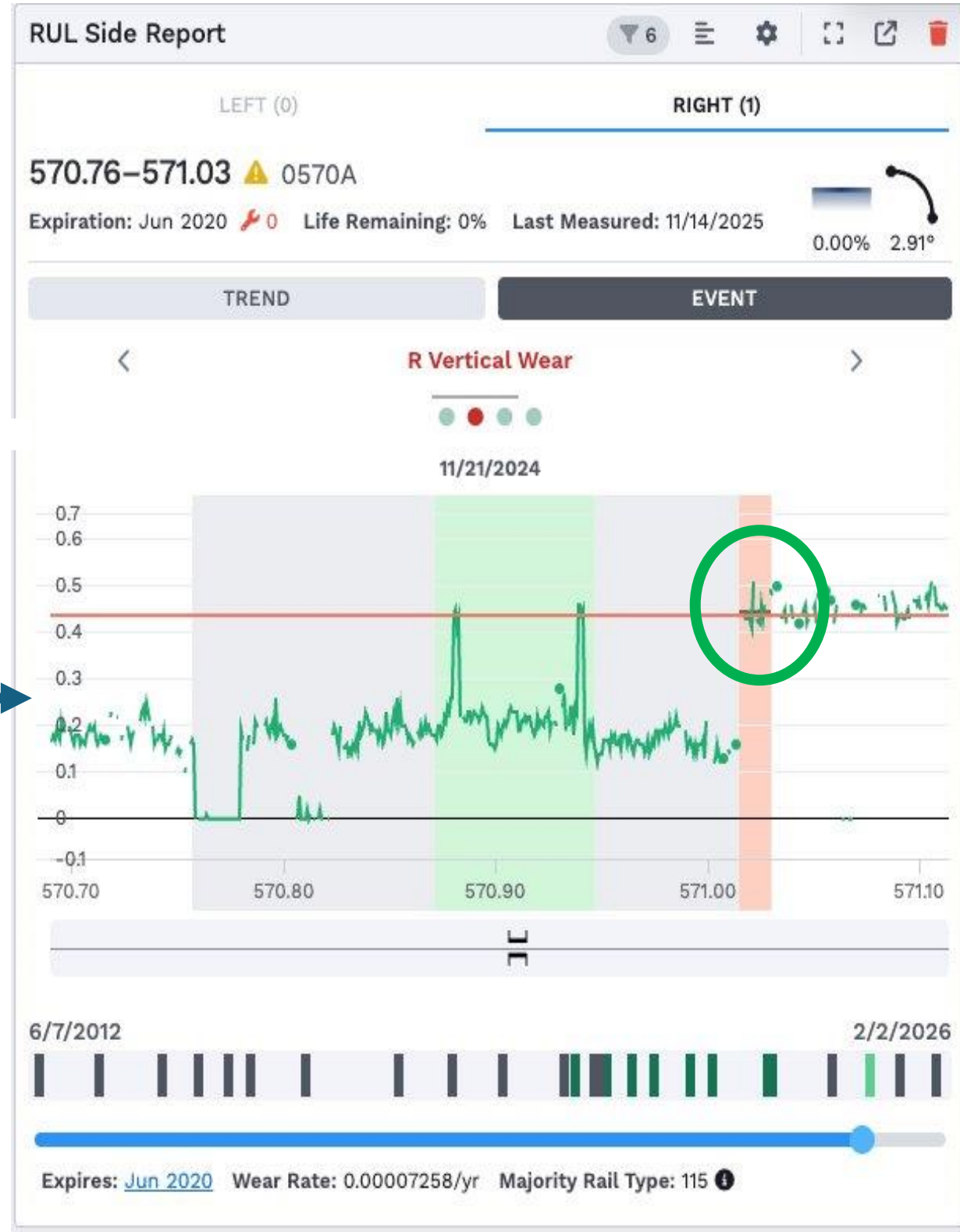
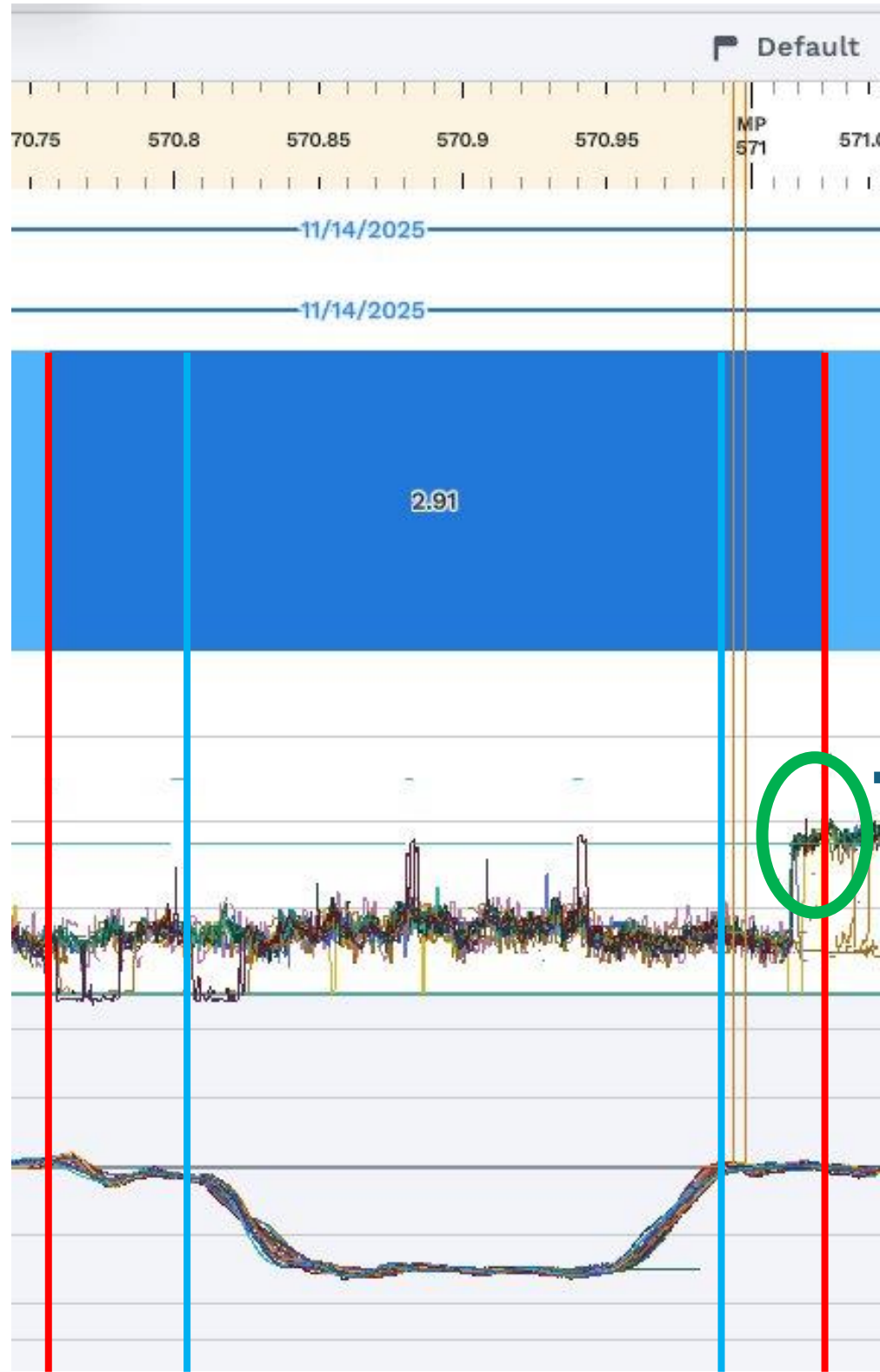
In one network of 25k curves there were nearly 2000 subparts within compound curves, each with its own degradation rate

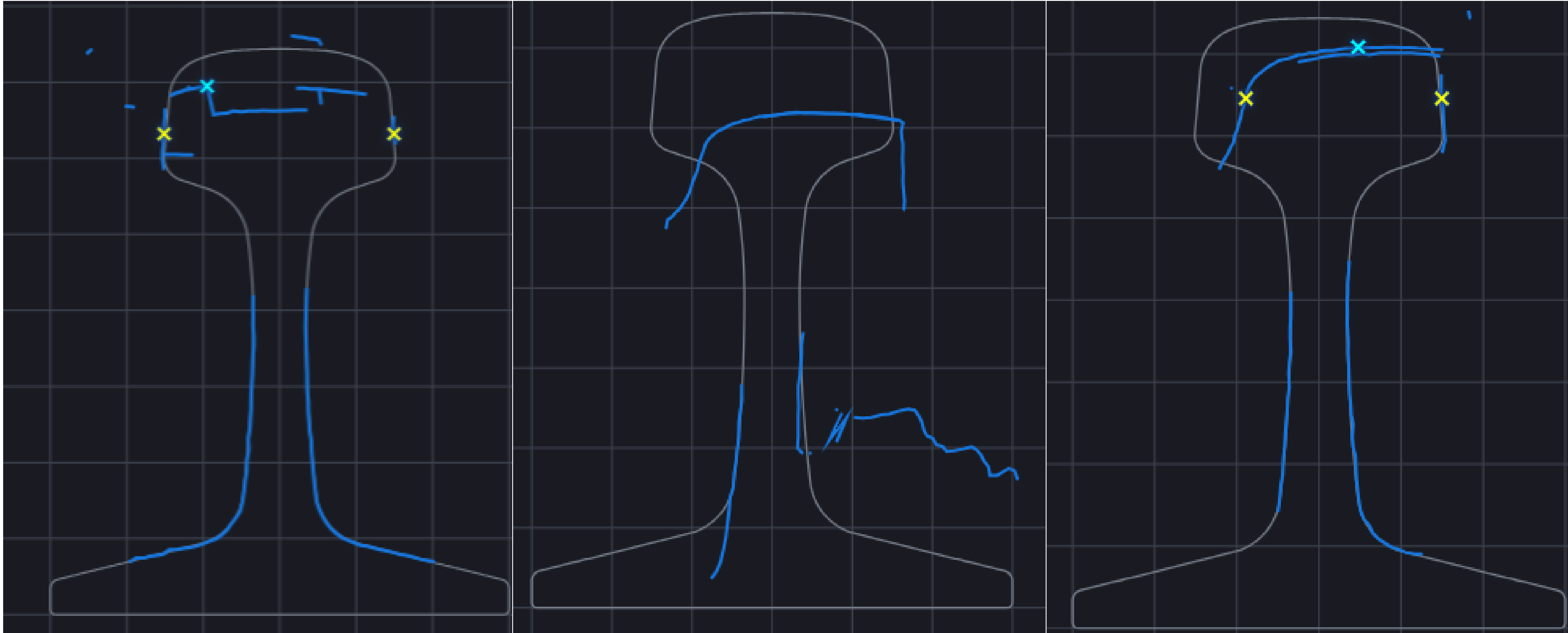




Asset Definition

Example: Data belonging to tangent influencing the predicted date



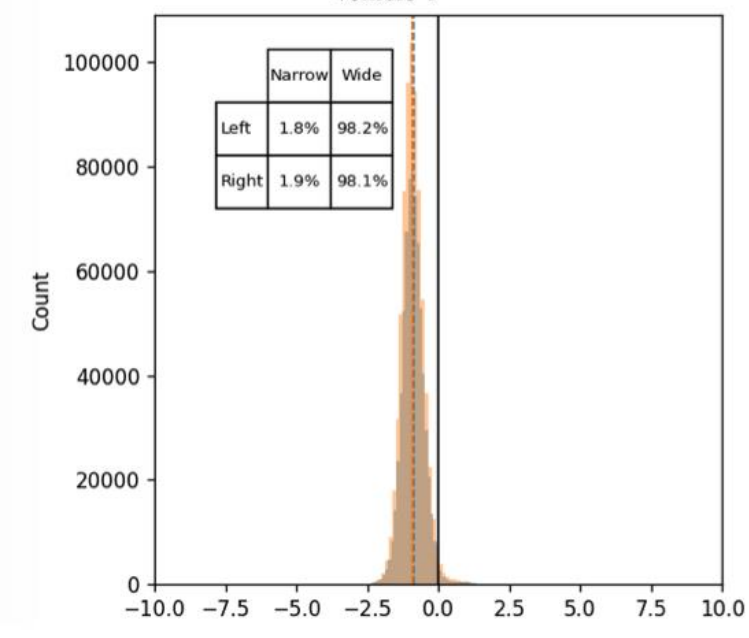




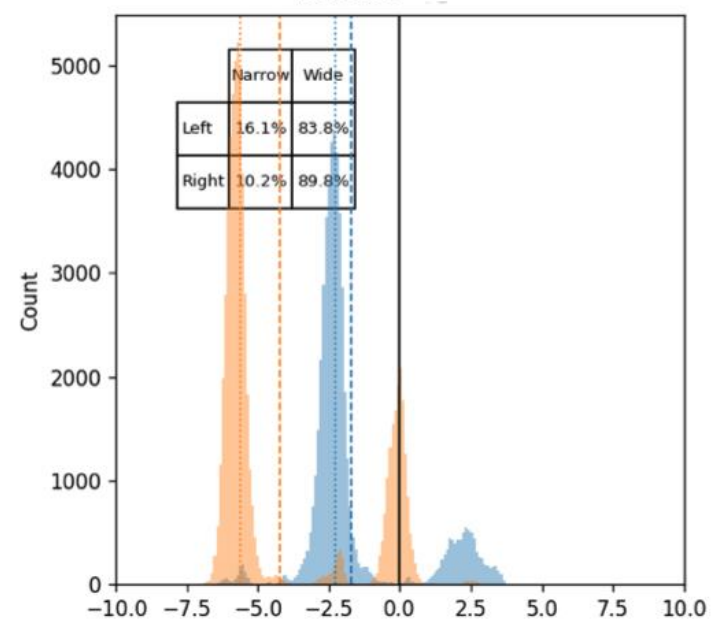
Data Collection



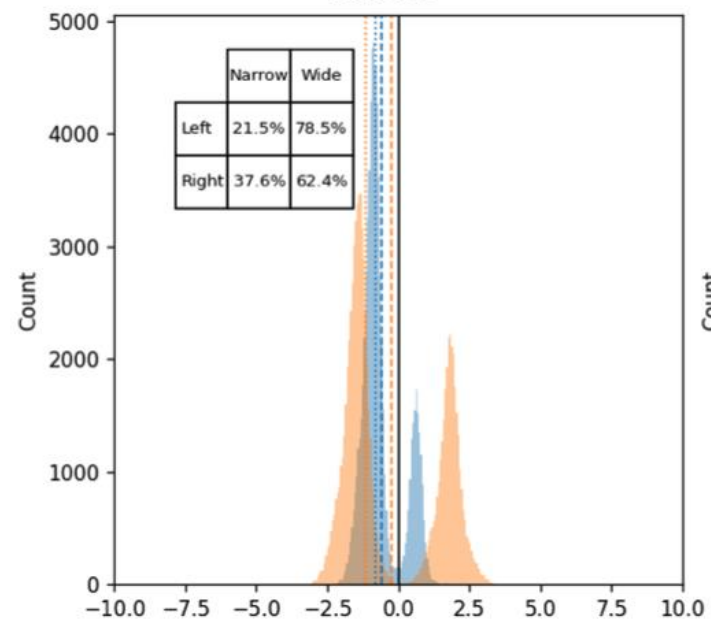
Vehicle 1



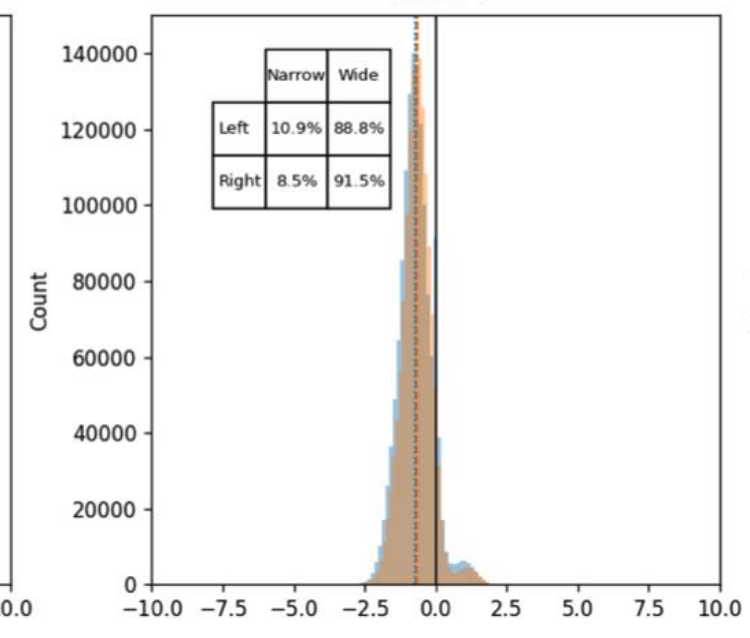
Vehicle 2



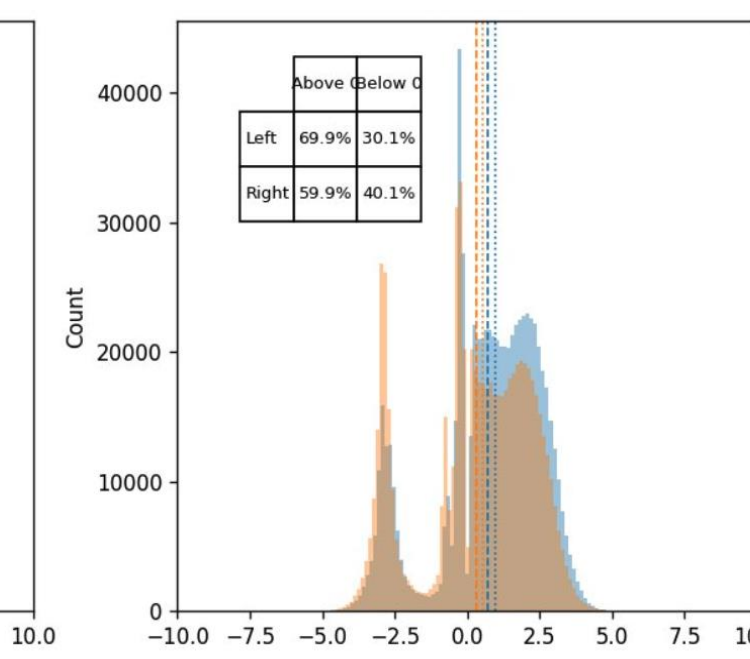
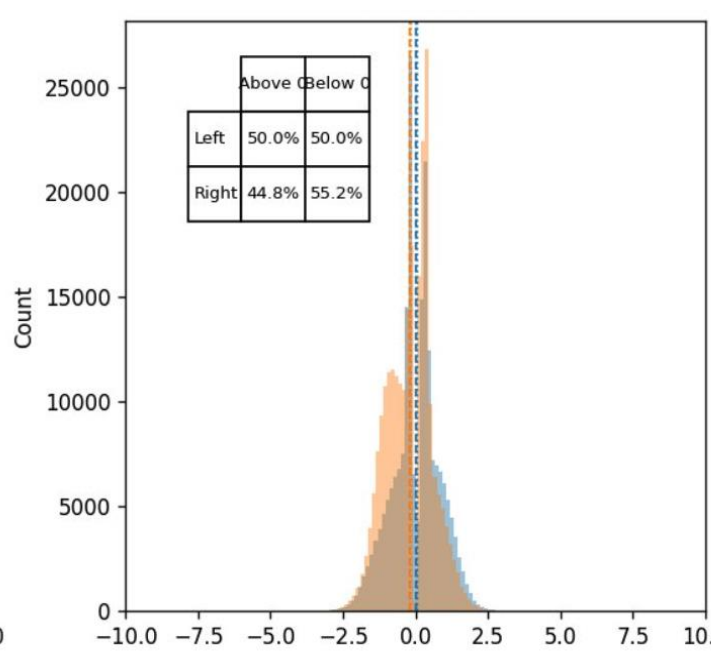
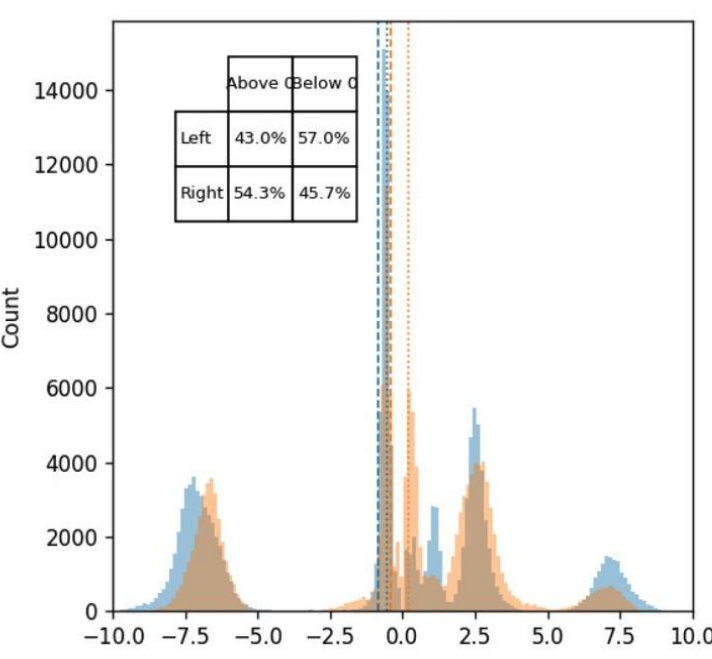
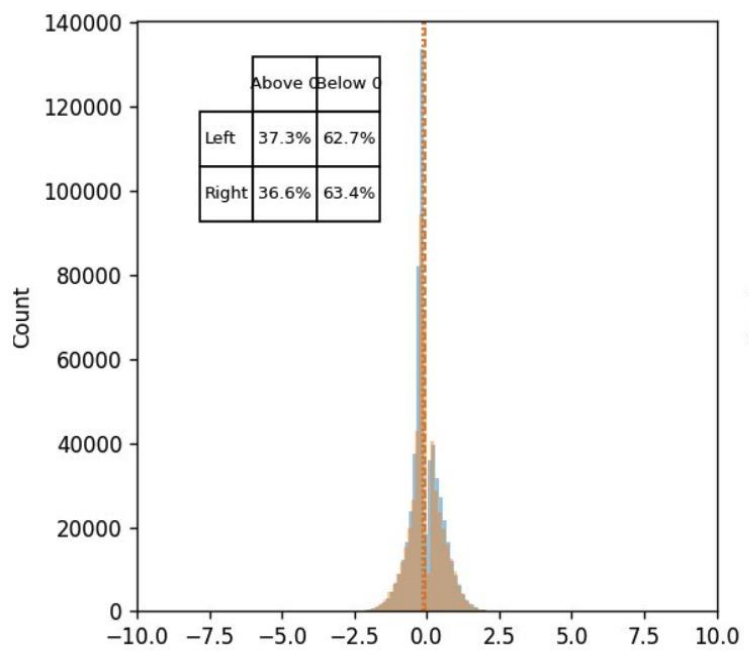
Vehicle 3



Vehicle 4



Horizontal Rail Profile Distortion

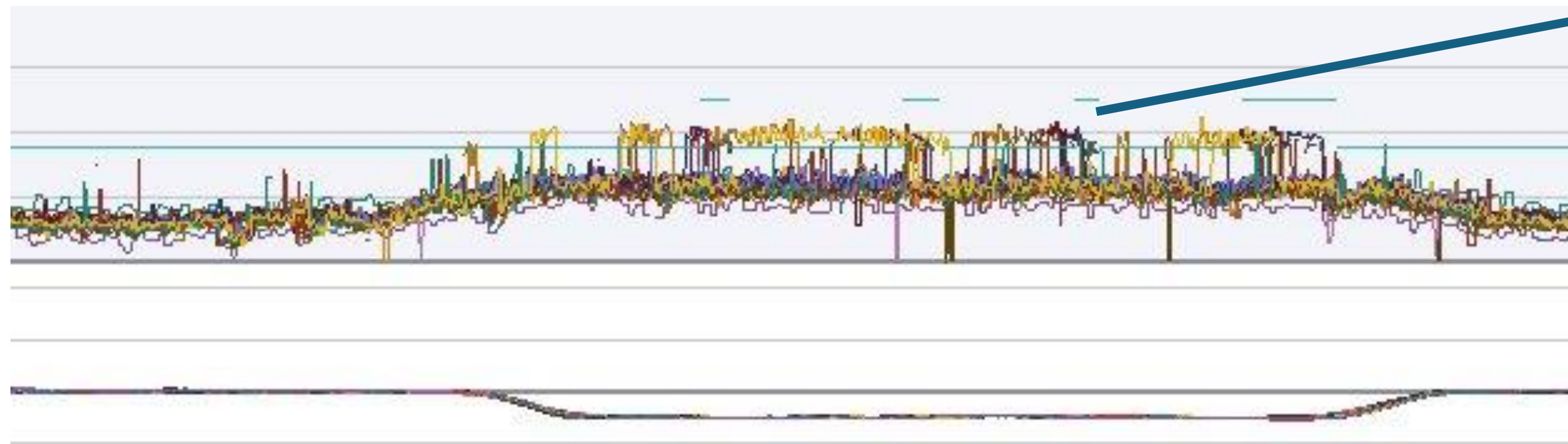


Vertical Rail Profile Distortion

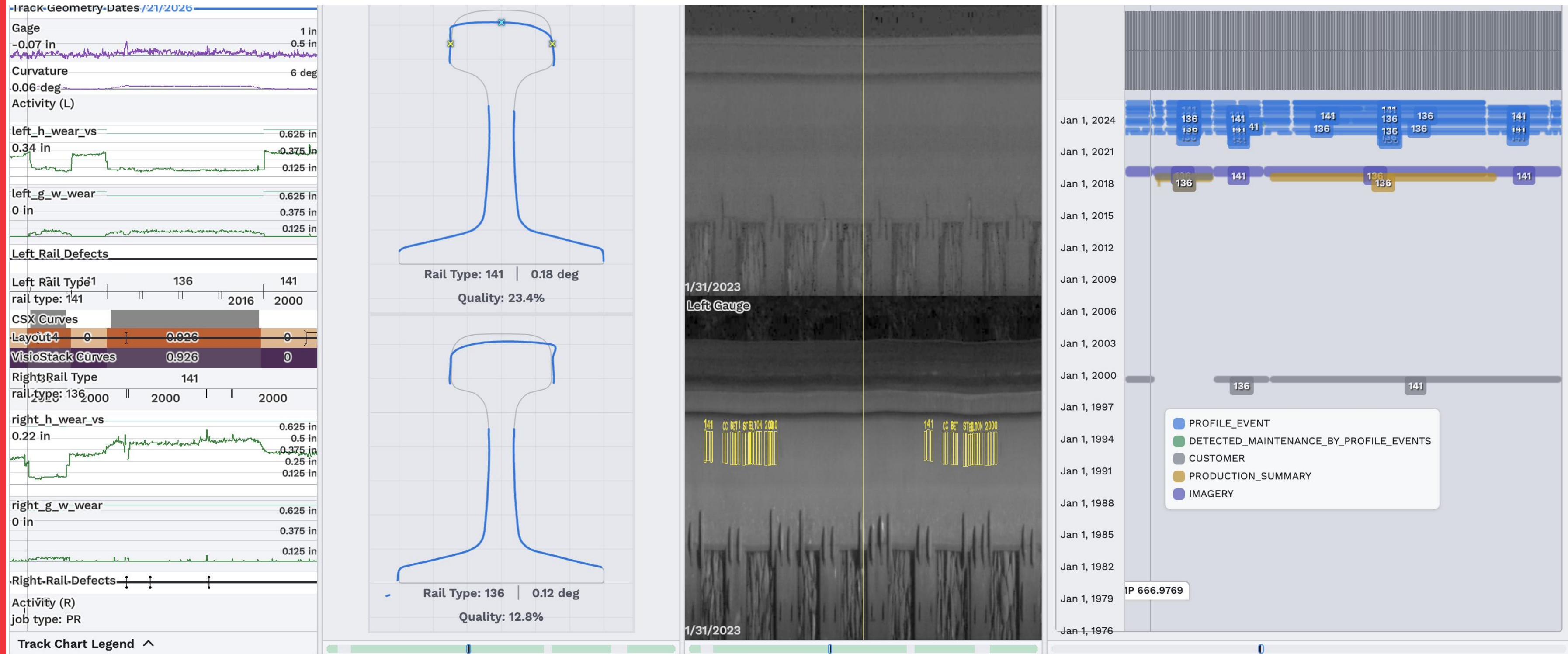


Incorrect rail size identification leads to inaccurate predictions

Predictions get unstable as points jump from type to type



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Swapped L/R data breaks trends and reduces confidence in predictions

Addressed in post processing with pattern recognition

285.55–285.69 ⚠️ 0285B

Expiration: Sep 2025 🔧 0

Life Remaining: 0%

Last Measured: 11/21/2025

0.05% 1.11°

TREND

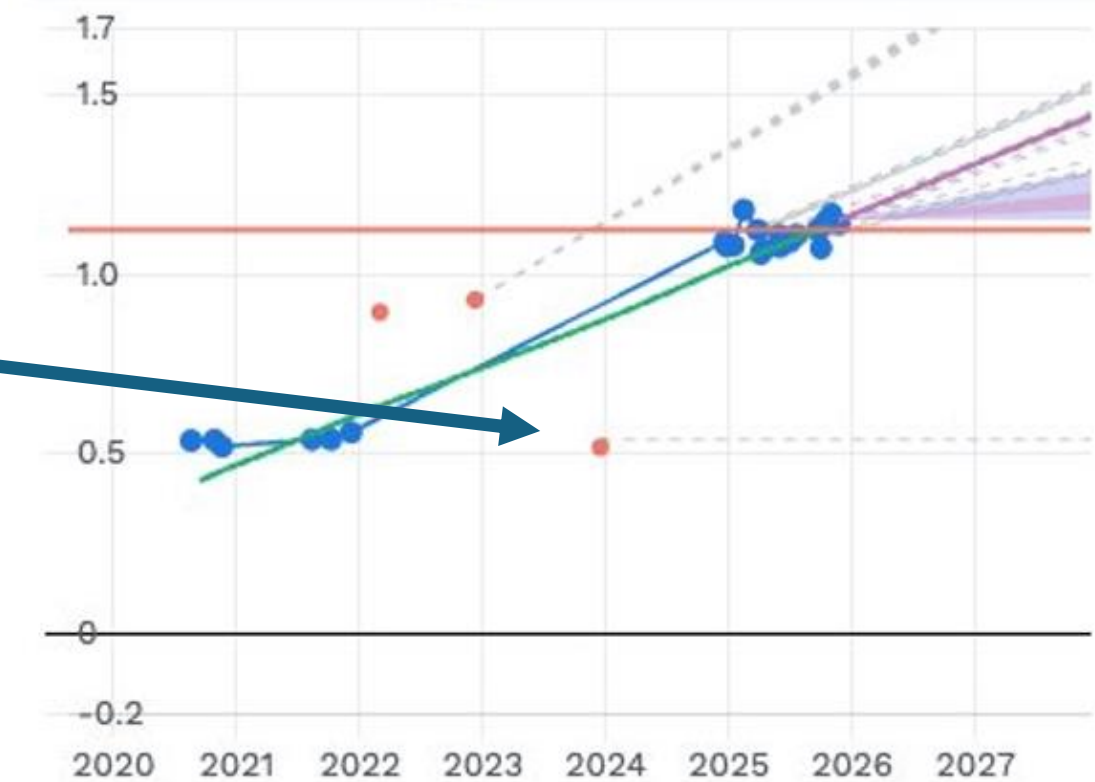
EVENT



RIGHT COMBINED WEAR



0

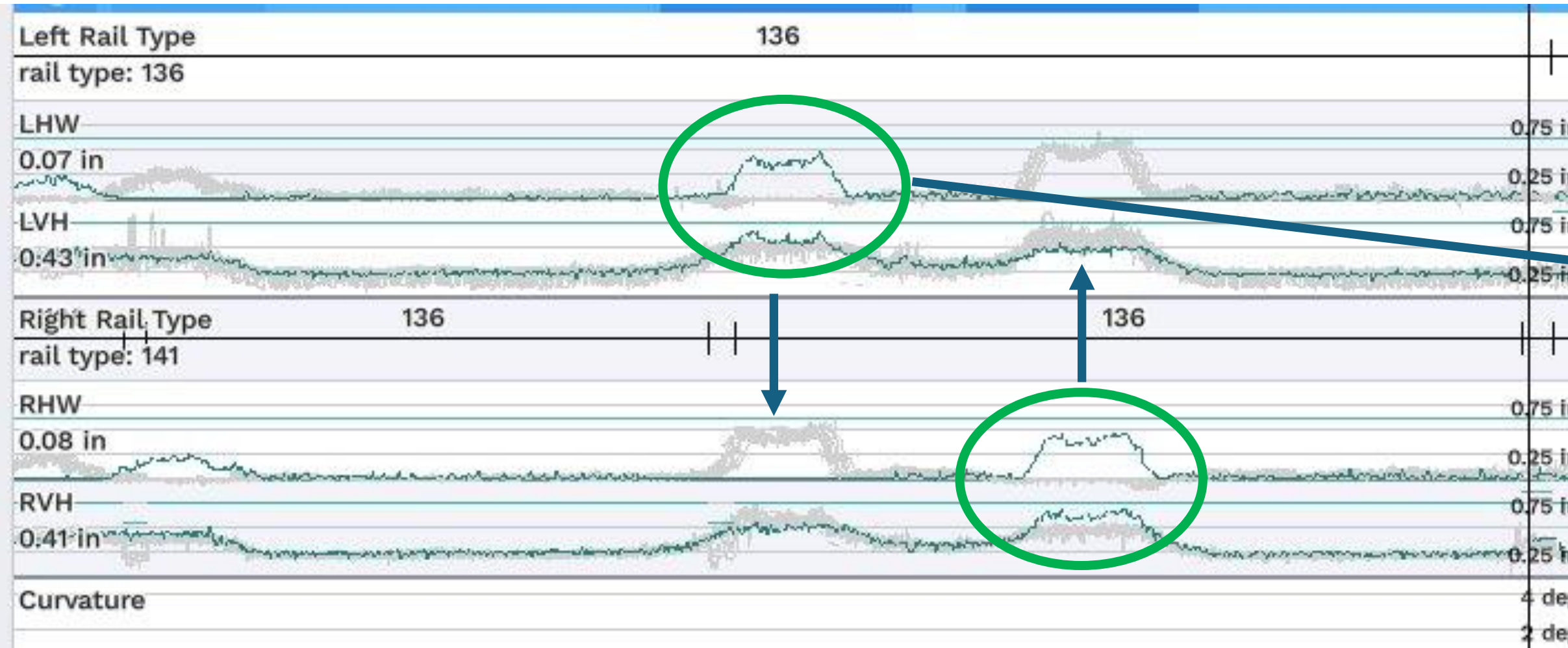


8/18/2019

12/9/2027

Tonnage 2.64058 4.81062

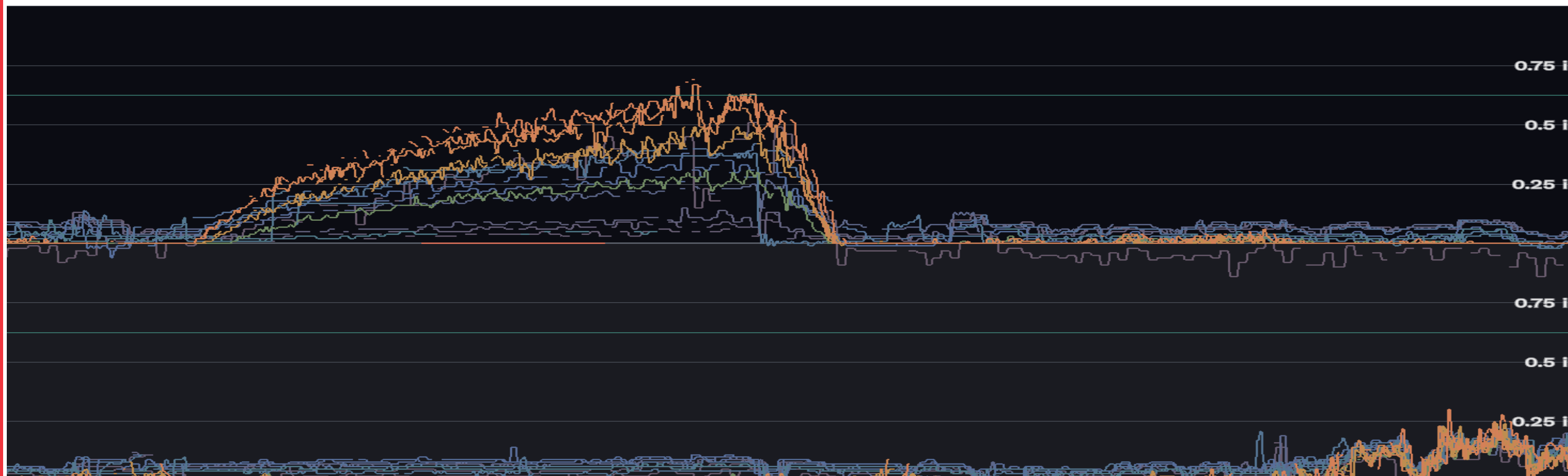
Expires: Sep 2025 Wear Rate: 0.1419/yr



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Before

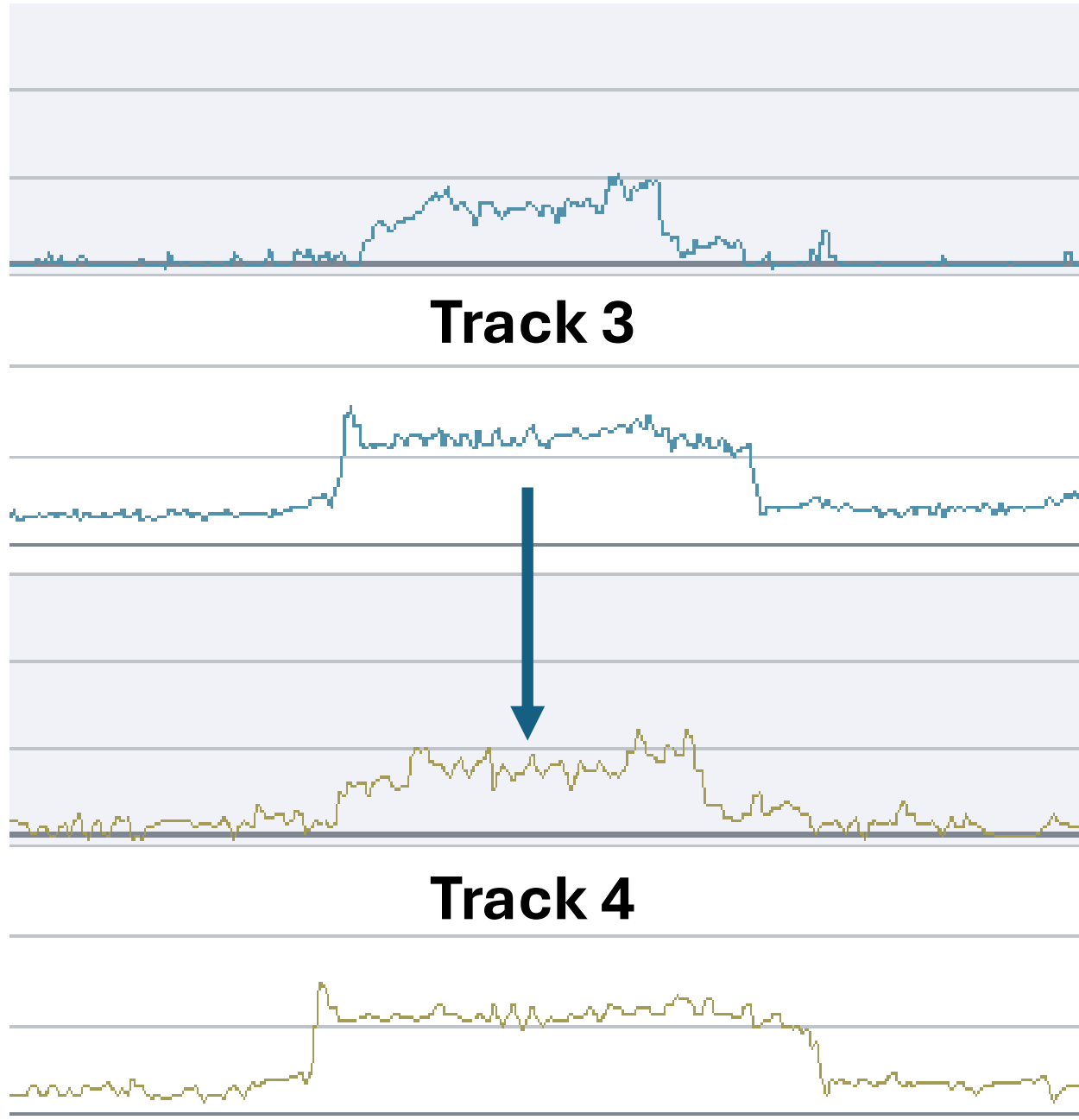


After

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Incorrect Tracks

Nearly 1,000 miles of data recovered on a single network



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Data Collection



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Expiration: Sep 2025 0

Life Remaining: 0%

Last Measured: 10/15/2025

TREND

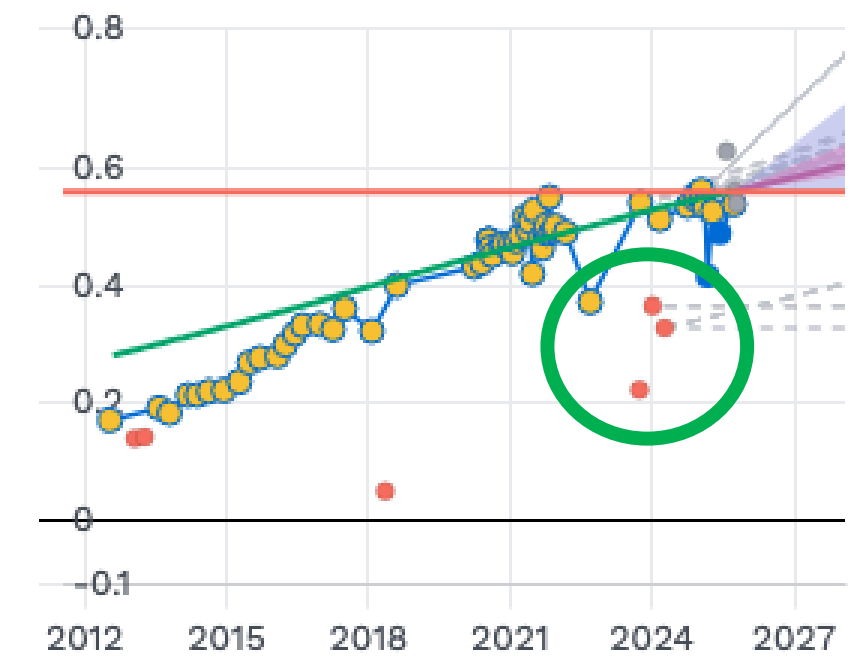
EVENT



R Vertical Wear



0



7/18/2011

1/28/2028

Tonnage 3.81984 8.29054

Expires: [Sep 2025](#)

Wear Rate: 0.01884/yr

Majority Rail Type: 132



Prediction

When a trend is reset and there's not much data to predict on, accurately forecasting can become a challenge

One tool at our disposal is to use historic cycles for the curve in question

With robust data history we can more accurately predict a replacement date by using the historic cycle durations to estimate the current cycle

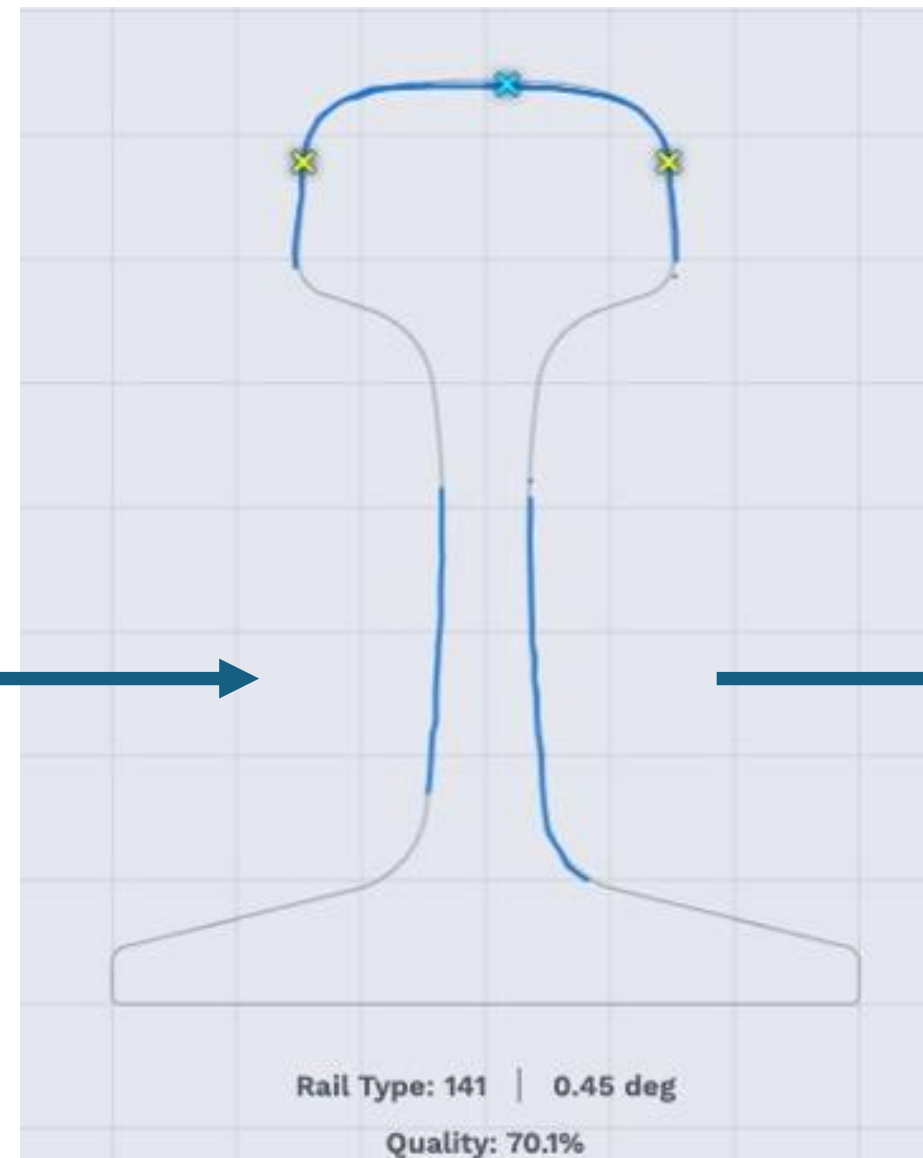
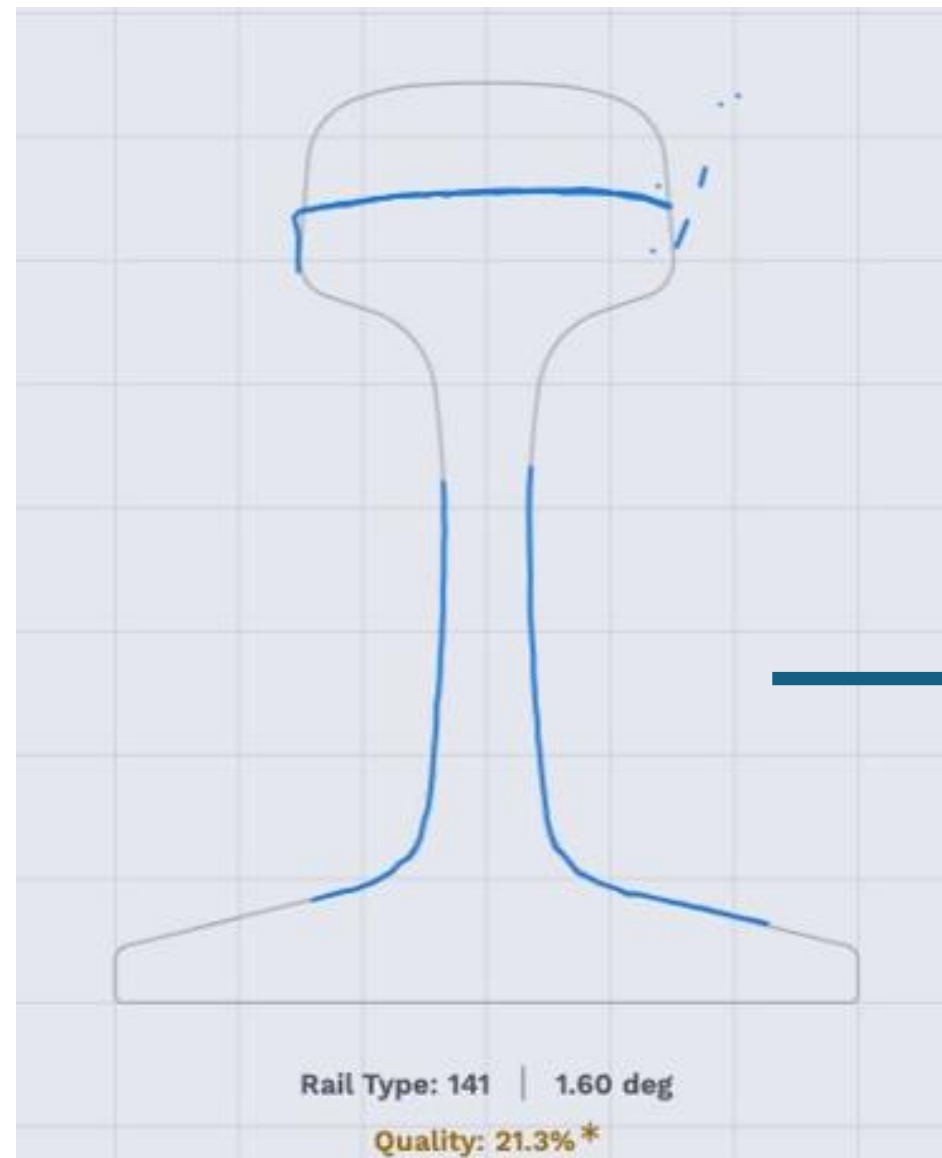




Prediction

Maintenance events represent a clear break in trending.

When maintenance data isn't available other methods of detection should be utilized



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Expiration: Oct 2024 2
Life Remaining: 0%
Last Measured: 11/10/2025

-0.64% 3.68°

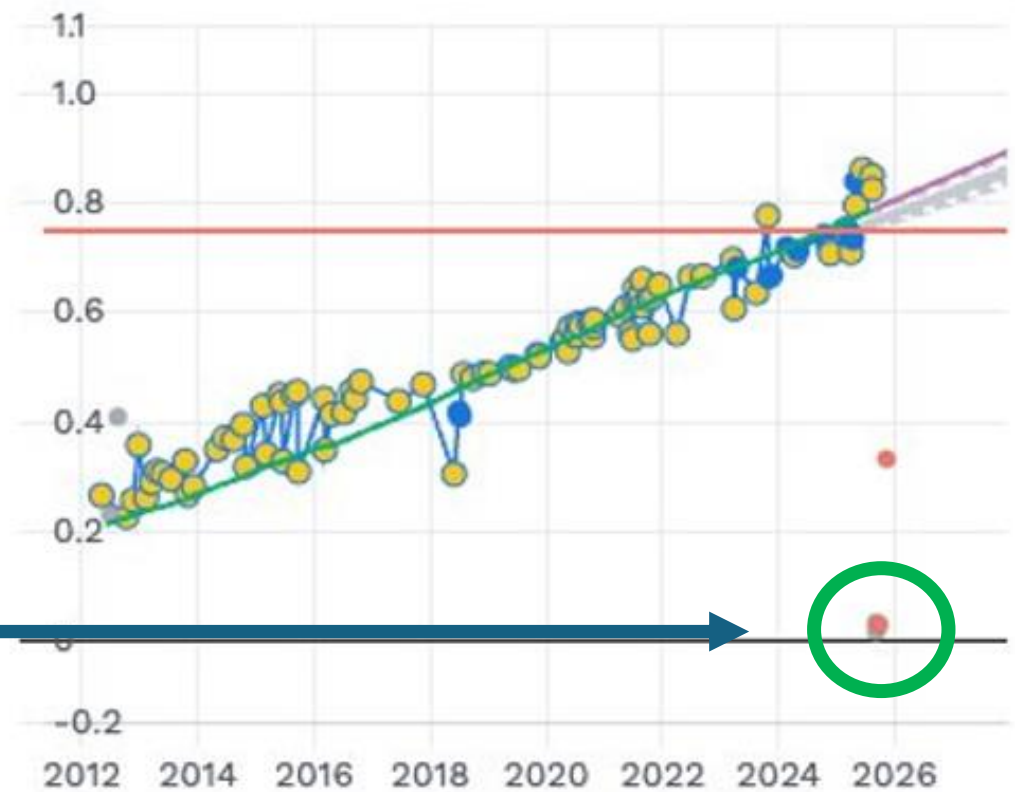
TREND

EVENT

L Vertical Wear



2



5/3/2011 12/8/2027

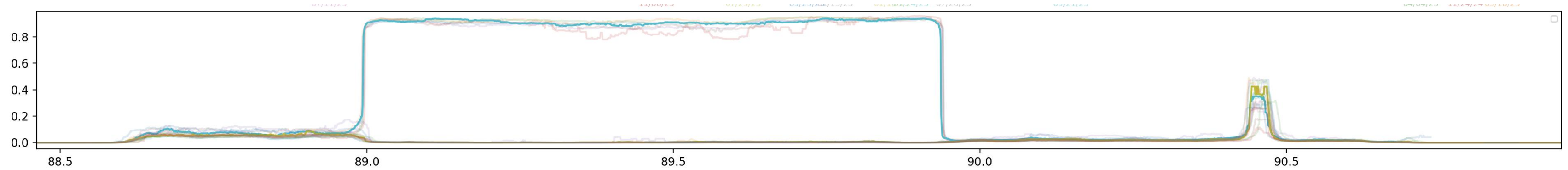
Tonnage 0.48928 13.19814

Expires: Oct 2024 Wear Rate: 0.04556/yr

Majority Rail Type: 136

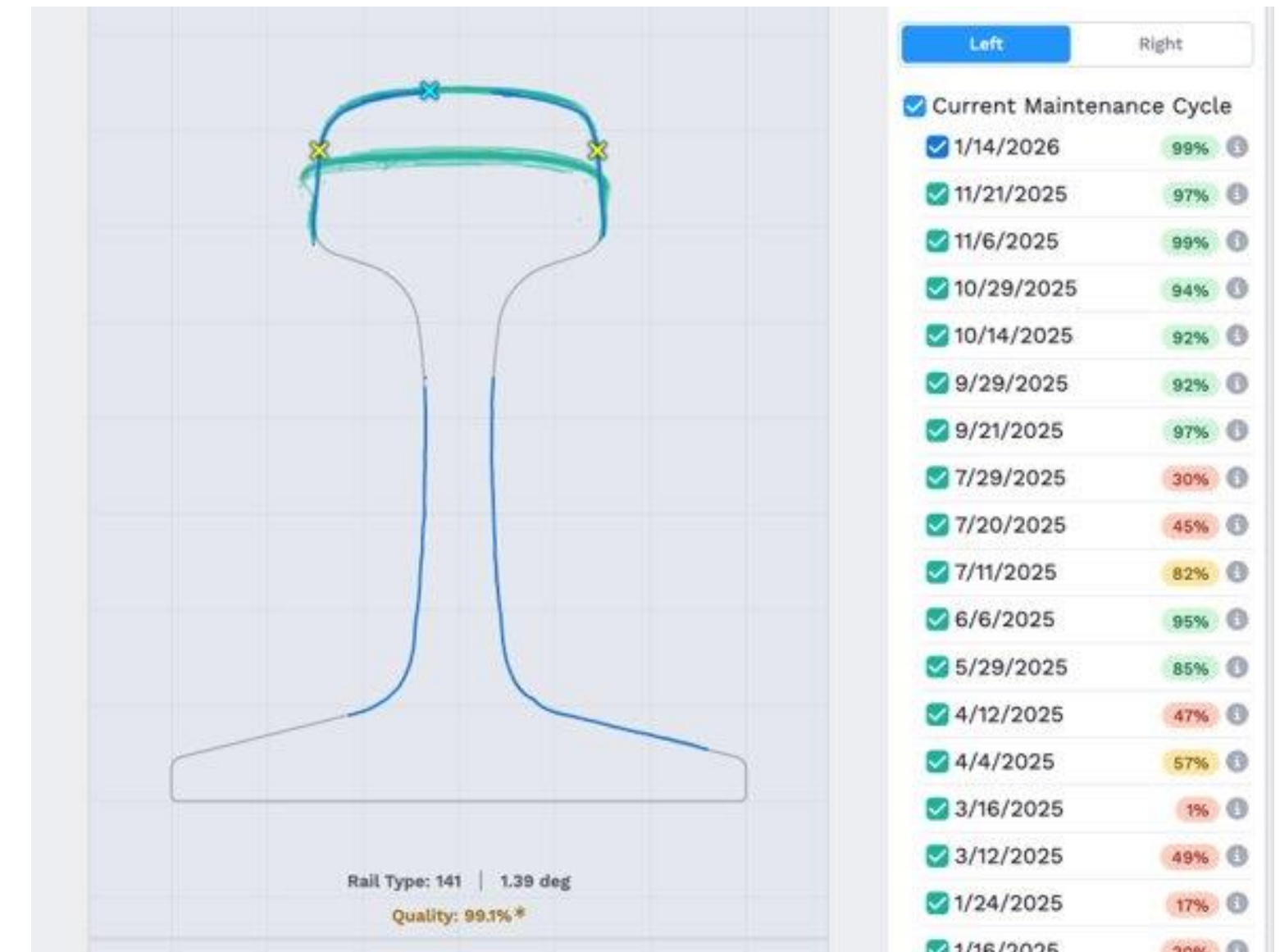


Prediction



The platform scores profiles based on worn characteristics.

When the platform detects a significant improvement in the score it will trigger a maintenance event detection

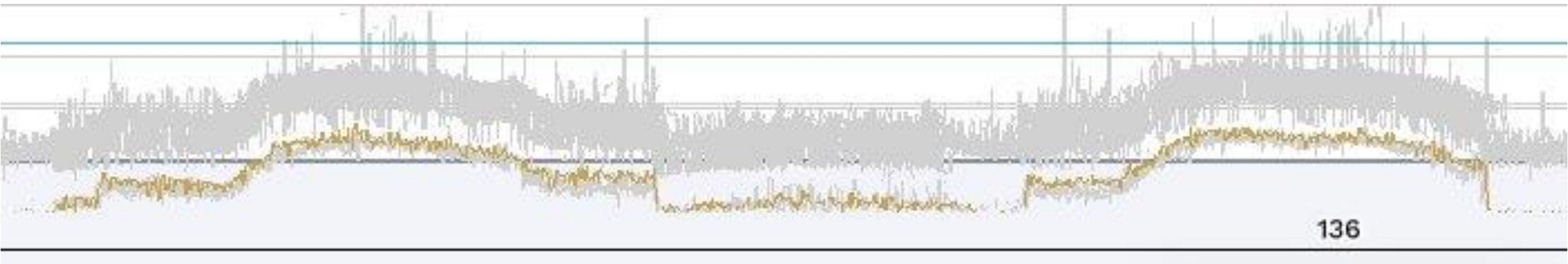


Prediction

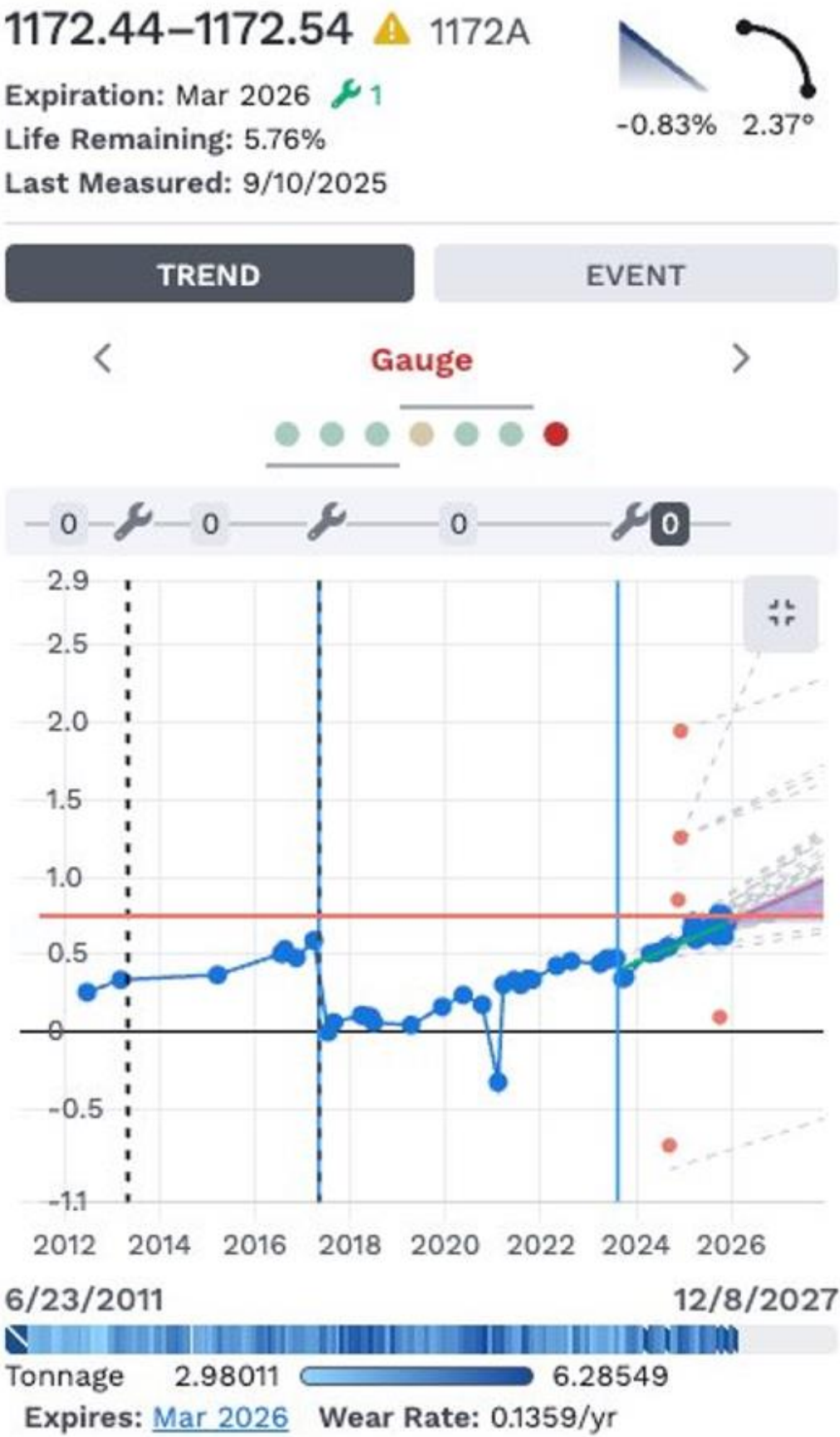
Outliers are a concern in any large data set

Within capital planning outliers can significantly change prediction dates leading to costly emergency relays

Addressing the root causes discussed in this presentation eliminates many cases where outliers could be created



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Prediction

Non-linear trends are unique cases where each curve behaves differently

- Does the tonnage vary over the life cycle
- Did lubrication of the curve change
- Has the metallurgy changed

Better understanding of non-linear trending can extend the forecast window out further without sacrificing accuracy

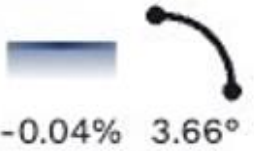


28.66–29.05 ⚠ 0028_

Expiration: Nov 2024 🔧 0

Life Remaining: 0%

Last Measured: 1/14/2026



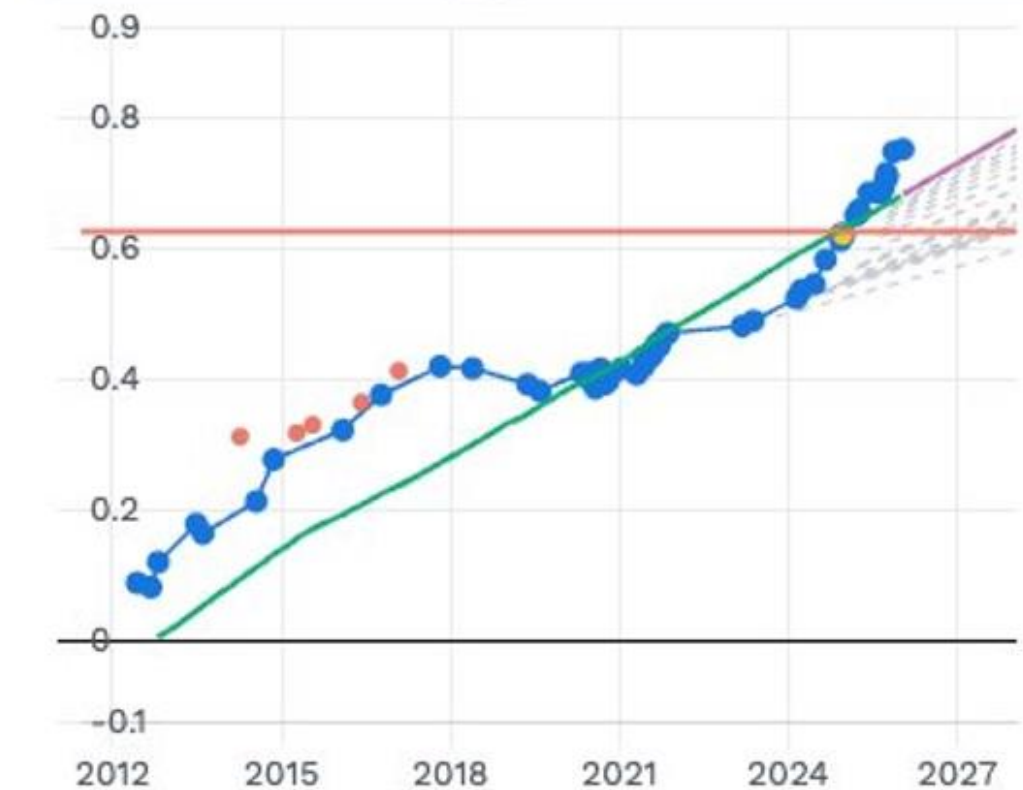
TREND

EVENT

< R Horizontal Wear >



9



6/9/2011

1/20/2028

Tonnage 1.31346 4.3411

Expires: Nov 2024 Wear Rate: 0.04931/yr

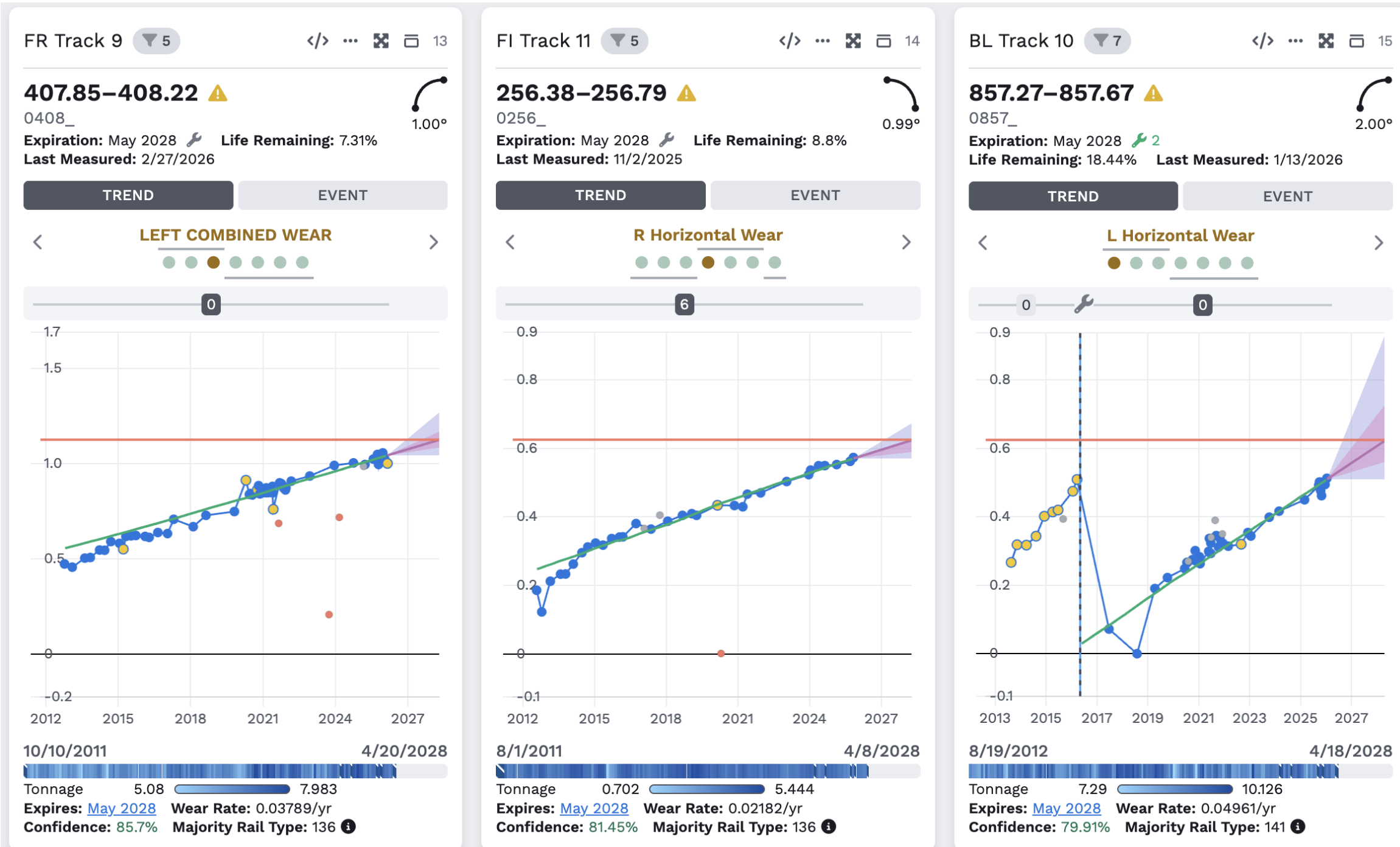
Majority Rail Type: 141 ⓘ

With so much data we can tell how a given curve should behave based on its characteristics

Looking at variables like degree of curvature, tonnage, grade, and others we can bucket like curves together

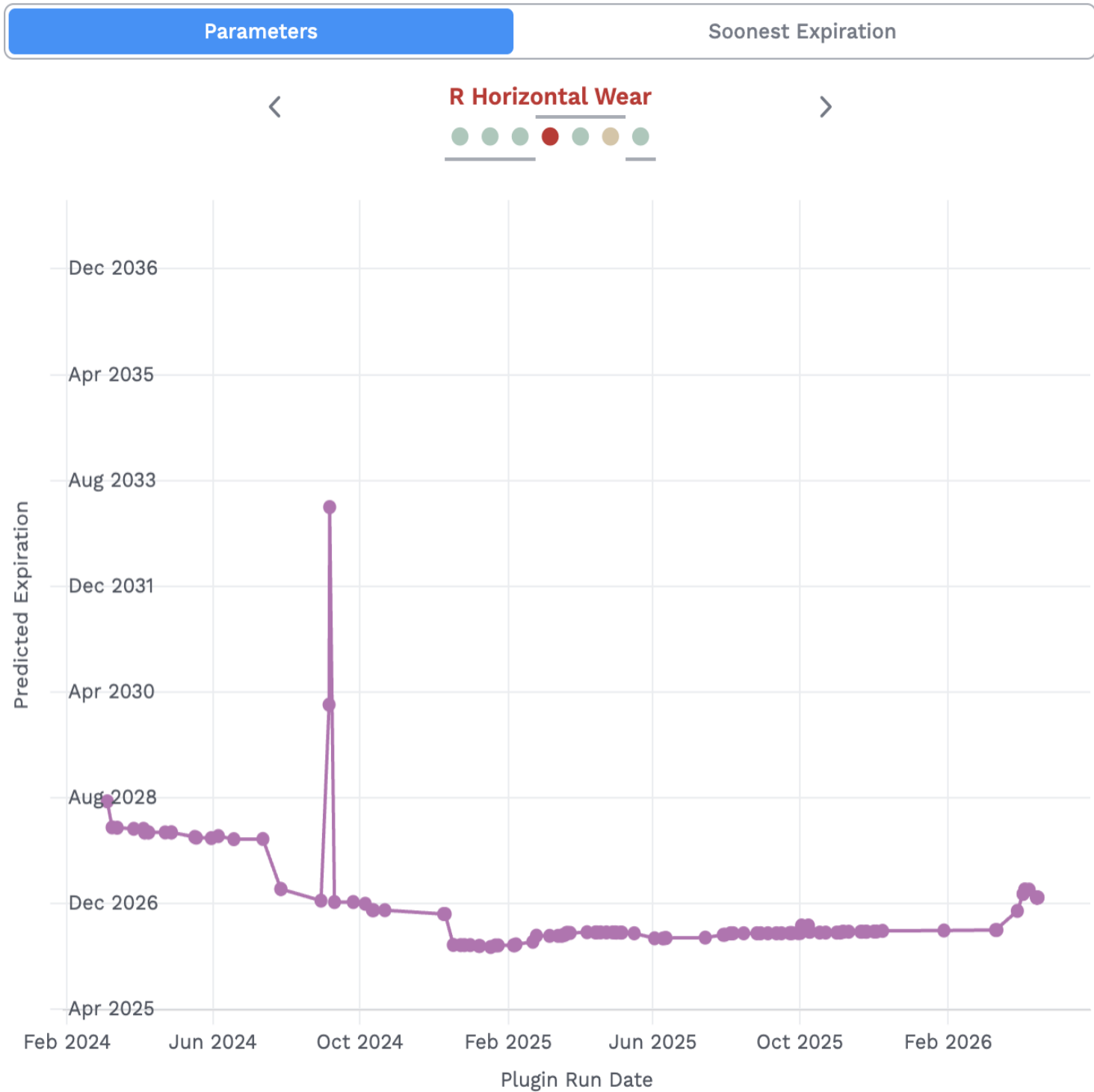
Probability fans let us call out where a curve is not behaving as we would expect

Identifies locations where the data may not be correct, or the curve wear pattern is behaving unusually





Prediction History



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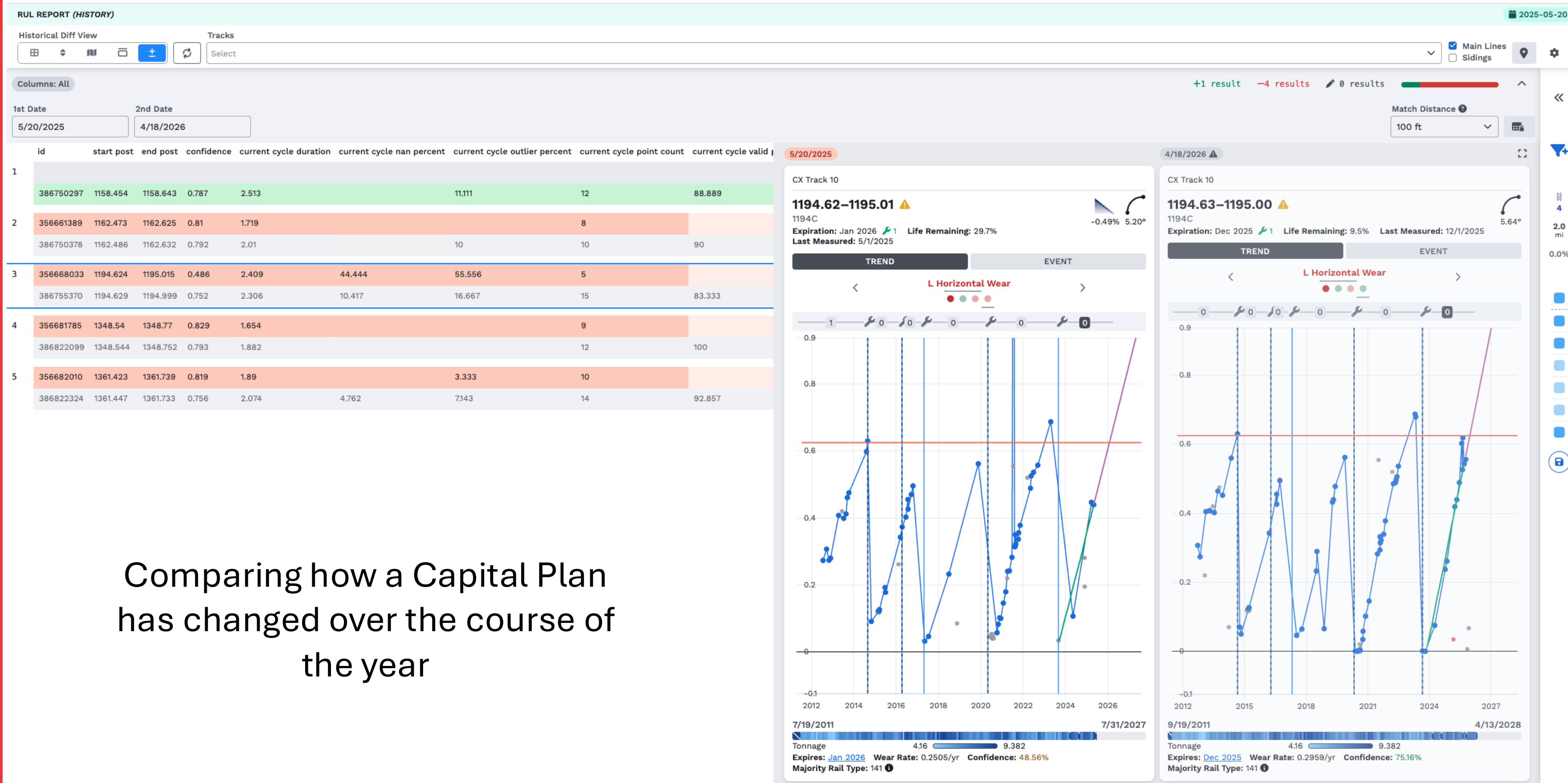
Being able to look back at what predictions looked like at any given time gives insights into the prediction

Especially useful for evaluating the stability of the prediction

Outliers and unusual predictions become very easy to identify in visual format



Comparing how a Capital Plan
has changed over the course of
the year



Combined RUL Graph

Parameter Trend

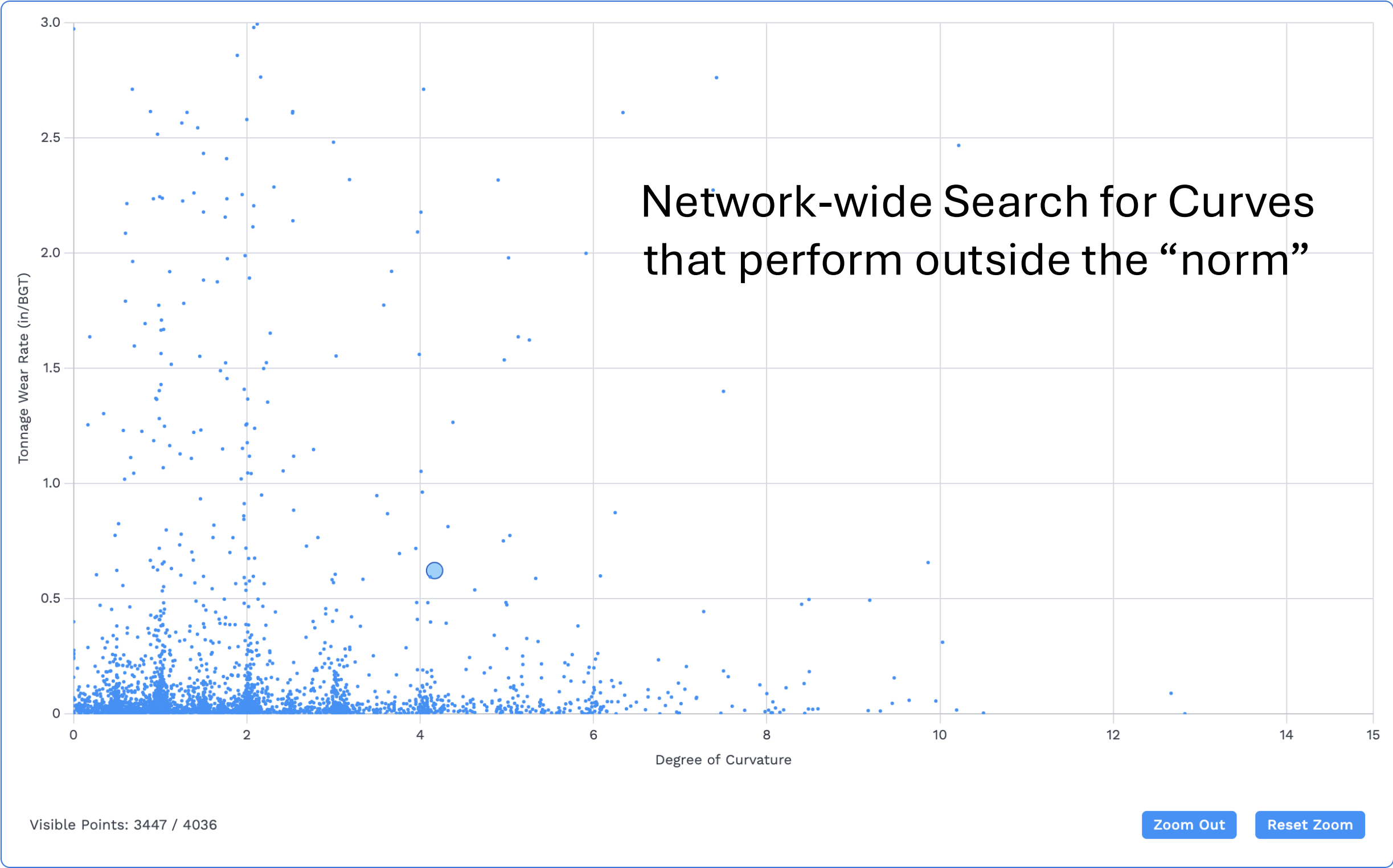
Low Horizontal Wear

Min Confidence Level

80

Max Confidence Level

100



BM Track 9

1646.82–1647.02

1647_

Expiration: undetermined

Life Remaining: 42.59%

Last Measured: 9/7/2025

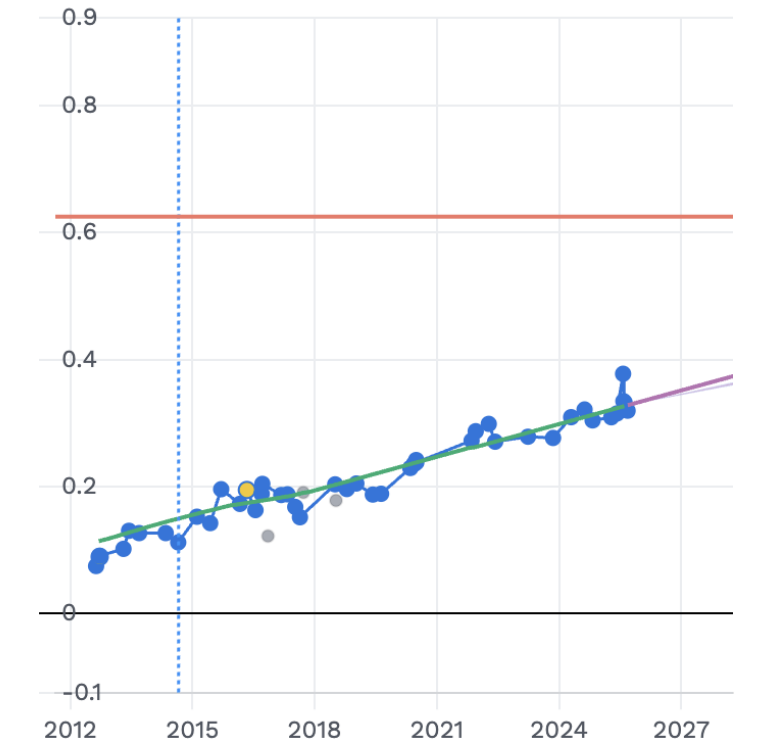
4.17°

TREND

EVENT

L Horizontal Wear

0



8/20/2011

4/8/2028

Tonnage

0.001

2.765

Expires: undetermined

Wear Rate: 0.01765/yr

Confidence: 81.9%

Majority Rail Type: 141

When it comes to capital planning there is no silver bullet that fixes everything

With time and investment in collecting good data capital planning doesn't have to be a year over year exercise

We believe a 3-5 year planning window is achievable with focus on the right variables

